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Overt visual attention modulates decision-related signals in ventral and dorsal medial prefrontal cortex

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eLife Assessment

This **important** work examines the effects of gaze on valuation signals in the human brain as participants choose between bundles of sequentially presented items food items. The paper provides **convincing** analyses of how gaze affects participants choice behaviour and how this varies across time. The work will be of interest to neuroscientists working on attention and decision-making.

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Abstract

When indicating a preference between two options, decision makers are thought to compare and accumulate evidence in an attention-guided process. Little is known about this process's neural substrates or how visual attention affects the representations of accumulated evidence. We conducted a simultaneous eye-tracking and fMRI experiment in which human subjects gradually learned about the value of two food-lotteries. With this design we were able to extend decisions over a prolonged time-course, manipulate the temporal onset of evidence, and therefore dissociate sampled and accumulated evidence. Consistent with past work, we found correlates of sampled evidence in ventromedial prefrontal cortex (vmPFC), and correlates of accumulated evidence in the prefrontal and parietal cortex. We also found that more gaze at an option increased its choice probability and that gaze amplified sampled-value signals in the vmPFC and ventral striatum. Finally, we found gaze modulated accumulated-value signals, above and beyond the non-gaze-modulated signals, in the pre-supplementary motor area (pre-SMA), providing novel evidence that visual attention has lasting effects on decision variables and suggesting that activity in the pre-SMA reflects accumulated evidence. These results shed new light on the neural mechanisms underlying gaze-driven decision processes.

Introduction

Many decisions involve a process of value computation and comparison between options. Imagine you are planning a vacation based on online travel-guides describing two locations. As you research these options, you sample pieces of information that support one option or another to varying degrees. For example, one destination might have cheaper flights while the other has cheaper hotels. Additionally, some information may attract more of your attention, such as pictures of waterfalls or ancient ruins. As you evaluate your options, you must integrate these pieces of information and eventually decide when to stop and make a final decision.

Decisions like this are thought to rely on a bounded, evidence-accumulation process that depends on factors such as the value of the sampled information and shifts in attention. According to this framework, when two options are similar in value, evidence accumulates more slowly towards the

decision threshold, resulting in longer response times (RT) and more opportunity for shifts in attention to influence the choice outcome. In contrast, when one option is clearly superior, evidence accumulates more rapidly and the decision is made quickly with less of a relation between gaze and choice. This choice process produces reliable, quantitative patterns in choice, RT, and eye-tracking data (Ashby et al., 2016 [↗](#); Callaway et al., 2021 [↗](#); Gluth et al., 2018 [↗](#); Krajbich et al., 2010 [↗](#); Smith & Krajbich, 2018 [↗](#)). For instance, decisions with similar values are more random (i.e., less predictable), tend to take more time (Konovalov & Krajbich, 2019 [↗](#)), and can be experimentally manipulated by diverting attention towards one option more than the other (Bhatnagar & Orquin, 2022 [↗](#); Gwinn et al., 2019 [↗](#); Pärnamets et al., 2015 [↗](#); Pleskac et al., 2022 [↗](#); Tavares et al., 2017 [↗](#)). Critically, these behavioral measures do not simply correlate; rather, they exhibit precise quantitative relationships consistent with evidence accumulation models (Konovalov & Krajbich, 2019 [↗](#)).

Sequential sampling models (SSMs) offer a framework to understand this evidence accumulation-driven choice process. SSMs vary in specific details, but generally share some core features. When faced with a decision, people begin to sample information in favor of each option. In many models, this information is evaluated and converted into relative evidence for one option or the other. Relative evidence builds up over time until there is enough to commit to an option (Busemeyer & Townsend, 1993 [↗](#); Ratcliff & Smith, 2004 [↗](#)).

Most SSMs can be broken down into two key components: inputs and integrators (Bogacz et al., 2006 [↗](#)). Inputs encode the drift rate – the rate of evidence accumulation – and integrators encode the decision variable – the amount of accumulated evidence.

Each option has its own input. In the context of value-based decisions, the input represents the value of the currently considered piece of information. For a given option, the average input value is generally assumed to be constant over the course of the decision but does vary randomly from one instant to the next due to stochasticity in the sampling process (Shadlen & Shohamy, 2016 [↗](#)).

Integrators accumulate the stochastic sequences of sampled input values. Typically, each option has its own integrator (Busemeyer & Townsend, 1993 [↗](#); Gold & Shadlen, 2007 [↗](#); Krajbich & Rangel, 2011 [↗](#); Usher & McClelland, 2001 [↗](#); Wang, 2002 [↗](#)) which accumulates the evidence from that option's input, but is also inhibited by the other options' inputs or integrators. Thus, each integrator represents the accumulated, relative evidence favoring a given option. Once one of the integrators' accumulated evidence reaches a pre-determined threshold, the corresponding option is chosen. In contrast to the input values, these accumulated values dynamically evolve over the course of the decision.

The neural inputs and accumulated values have been successfully identified in perceptual decision making (for recent reviews see: Forstmann et al., 2016 [↗](#); Hanks & Summerfield, 2017 [↗](#); O'Connell et al., 2018 [↗](#); Ratcliff et al., 2016 [↗](#)), but it has proven more challenging for value-based decisions. The main reason is that decisions are typically very quick, with RTs shorter than the time resolution of functional magnetic resonance imaging (fMRI) (but see Gluth et al., 2012 [↗](#)). Evidence for the neural substrates of value-based SSMs have typically come as trial-level measures correlating with model parameters (Hare et al., 2011 [↗](#); Rodriguez et al., 2015 [↗](#)), scalp-level electric activity from electroencephalography (EEG) (Polanía et al., 2014 [↗](#)), or a combination of the two (Pisauro et al., 2017 [↗](#)). Taken together, these studies have implicated a fronto-parietal network underlying value-based decision-making, with more ventral/frontal regions serving as inputs (for reviews see: Bartra et al., 2013 [↗](#); Clithero, 2018 [↗](#)) and more dorsal/parietal regions serving as integrators.

Despite their many strengths, these past value-based experiments have been limited by their inability to determine whether purported integrator regions are accumulating evidence or instead representing unchosen values (Boorman et al., 2011 [↗](#); Kolling et al., 2016 [↗](#); Wittmann et al., 2016 [↗](#)), decision conflict (Frömer et al., 2019 [↗](#); Kaanders et al., 2021 [↗](#); Shenhav et al., 2014 [↗](#), 2016 [↗](#); Vassena et al., 2020 [↗](#)), or time on task (Grinband et al., 2011 [↗](#); Holroyd et al., 2018 [↗](#); Mumford et al., 2024 [↗](#)). In most experiments, these variables are highly correlated and difficult to

distinguish. Increasing the value of the worse option while holding the better option constant will simultaneously increase the perceived conflict, increase the deliberation time, and slow the rate of evidence accumulation.

To distinguish between accumulated evidence and the other confounding explanations, we sought a factor that modulates accumulated evidence within a decision, independent of time. For this we turned to visual attention, measured with eye-tracking. Research on the attentional drift diffusion model (aDDM) has argued that gaze amplifies value during the choice process (Bhatnagar & Orquin, 2022 [↗](#); Krajbich et al., 2010 [↗](#); Sepulveda et al., 2020 [↗](#); Smith & Krajbich, 2019 [↗](#); Westbrook et al., 2020 [↗](#)). This means that current gaze location should amplify value signals in the input regions, and that the balance of gaze allocation over the course of the decision should amplify accumulated evidence signals in the integrator regions. Both human and non-human primate research has confirmed gaze effects on value inputs in the orbitofrontal cortex (Lim et al., 2011 [↗](#); McGinty et al., 2016 [↗](#); Rich & Wallis, 2016 [↗](#); Hunt et al., 2018 [↗](#); but see McGinty, 2019 [↗](#)). However, it has yet to be shown that these gaze-modulated inputs are integrated into accumulated decision values. Gaze modulated signals in purported integration regions would provide critical evidence against the alternative explanations (i.e., conflict, time, or unchosen value).

Here, we present the results of an fMRI experiment designed to provide evidence that integrator regions accumulate gaze-weighted evidence. Our approach was to slow down the decision process by gradually presenting choice-relevant information. Our task design allowed us to extend the decision-making period to approximately a minute, while also allowing us to dissociate the inputs' sampled value (SV) signals from the integrators' accumulated value (AV) signals (Gwinn et al., 2019 [↗](#)). The inputs represent the perceived value of stimuli currently on the screen, while the integrators represent the values of previously presented stimuli within that choice problem. We simultaneously collected eye-tracking data, allowing us to test whether gaze modulates SV and AV representations. We specifically focus on the difference in sampled values between the left and right options (ΔSV) and the corresponding difference in accumulated values between the left and right options (ΔAV). We hypothesized that we would find (1) a positive correlation between gaze-weighted $|\Delta SV|$ and activity in the reward network (the ventromedial prefrontal cortex (vmPFC) and ventral striatum), and (2) a positive correlation between gaze-weighted $|\Delta AV|$ in the pre-supplementary motor area (pre-SMA) (Aquino et al., 2023 [↗](#)), dorsolateral prefrontal cortex (dlPFC), and intraparietal sulcus (IPS).

Results

Experiment description

Our choice task builds on an extensive literature examining choices between familiar snack foods. Instead of choosing between two food items, which typically only takes a few seconds, we asked subjects to choose between two food lotteries. A lottery consisted of 3-6 different items, each with a different probability of being selected. Subjects did not know anything else about the lotteries; they had to learn about them from experience (Hertwig et al., 2004). Specifically, subjects sampled a random draw from both lotteries every 4-8 seconds. They continued to sample random draws until they were ready to stop and choose one of the two lotteries (Fig. 1 [↗](#)). Choosing a lottery led to a final random draw from that lottery, revealing the actual food that the subject would receive if that trial was rewarded at the end of the experiment.

We placed no explicit limit on the number of draws subjects could sample within each trial (i.e. pair of lotteries). To prevent subjects from spending the entire session on a single trial, we gave them 45 minutes to make at least 60 choices. Subjects were informed that any unmade choices would be randomly completed by the computer, and any trials beyond 60 would be added to the list from which the rewarded trial would be drawn. If a subject were to sample the same number of stimuli in each trial, the optimal number would be seven. However, subjects could (and did)

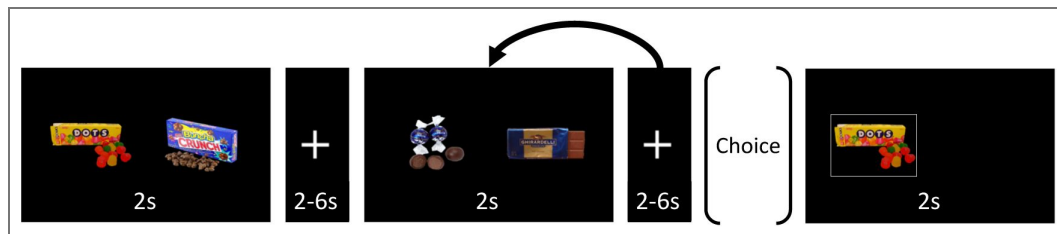


Figure 1. Task timeline.

Subjects chose between two snack food lotteries on each trial. Subjects learned about the lotteries through random food draws. Every 4-8 seconds, subjects sampled a new draw from each lottery. They were allowed to sample as many times as they wanted but were incentivized to sample approximately 7 draws per trial. Sampled food draws were presented for 2 seconds, followed by a fixation cross appearing for 2-6 seconds with random jitter. The trial ended when the subject chose the left or right lottery, using the respective index finger. Upon making their choice, subjects were presented with a food drawn from their chosen lottery.

vary their number of samples throughout the experiment; we observed substantial variability in the number of samples per trial within most of our subjects (mean number of samples = 6.37 and mean within-subject $SD = 2.61$).

We constructed each trial's sequence of items pseudo-randomly to minimize the correlation between $|\Delta SV|$ (i.e., the absolute difference between the left and right values of the currently presented stimuli) and $|\Delta AV|$ (i.e., the absolute sum of ΔSV at a given time within a trial) (see Methods). For the first sample in each trial, ΔSV and ΔAV were always equal. After subsequent samples, ΔSV diverged from ΔAV , yielding two distinct time courses to look for in the fMRI data (Fig. 2 [↗](#)).

To measure the subjective value of each stimulus, we separately asked subjects to rate all the food items. Before entering the scanner, subjects rated 148 unique food items based on how much they would like to eat them at the end of the experiment. These ratings were incentivized (see Methods) and we retained only the positively rated items (0 to 10) for the choice task. We used each subject's ratings to calculate SV and AV.

Behavioral results

A core assumption of SSMs is that individuals decide based on the evidence accumulated over the course of the decision. We thus anticipated that subjects would choose in line with ΔAV and not just the most recent ΔSV in the trial. We tested this key assumption with a mixed-effects logistic regression of choosing the left lottery on ΔSV and ΔAV at the time of choice. Subjects chose in line with both (ΔAV excluding the final samples: $\beta = 0.062$, $SE = 0.010$, $p < 0.001$; ΔSV : $\beta = 0.257$, $SE = 0.024$, $p < 0.001$) (Fig. 3A [↗](#)).

The larger coefficient for ΔSV compared to ΔAV is an inevitable consequence of an SSM choice process. In SSMs, a choice occurs when accumulated evidence reaches a threshold. Critically, perceived value for any given sample consists of the true underlying value plus random noise. The final sample (ΔSV) is what pushes the accumulated evidence over the threshold, which creates a selection effect: decisions tend to be made when the noise component of ΔSV is relatively large and aligned with the ultimate choice, causing the perceived final ΔSV to systematically overestimate the true ΔSV . As a result, the regression coefficient for the effect of final ΔSV on choice is overestimated. In contrast, ΔAV represents the sum of all previous evidence, which includes samples that were insufficient to trigger a choice and thus more likely to have noise components that favored the non-chosen option. This means that the perceived ΔAV systematically underestimates the true ΔAV . As a result, the regression coefficient for the effect of ΔAV on choice is underestimated. This creates an inherent asymmetry between ΔSV and ΔAV : even when the true decision process weights evidence equally over time, regression analyses will show larger coefficients for ΔSV than ΔAV . For any data generated by an SSM, regressing choice probability on final ΔSV and total ΔAV would produce a larger coefficient for ΔSV due to this threshold-crossing selection effect.

A second feature of SSM data is that easier decisions generally take less time (i.e., fewer samples). Therefore, we expected a negative correlation between the number of samples and the absolute difference in expected value between the two lotteries, as well as a higher probability of terminating a trial when the absolute value difference ($|\Delta AV|$) is higher. A mixed-effects regression of $\log(n \text{ samples})$ on the absolute expected value difference between the two lotteries revealed a negative relationship ($\beta = -0.025$, $SE = 0.009$, $p = 0.006$, two-sided t-test). A mixed-effects logistic regression of $P(\text{stop sampling})$ on $|\Delta AV|$ also revealed a significant positive relationship ($\beta = 0.062$, $SE = 0.009$, $p < 0.001$, two-sided t-test). These tests confirm that our subjects were sampling more in more difficult trials.

A third behavioral pattern predicted by the aDDM and other gaze-based SSMs is that individuals should generally choose options that they have looked at more (Cavanagh et al., 2014 [↗](#); Krajcich et al., 2010 [↗](#); Shimojo et al., 2003 [↗](#); Thomas et al., 2019 [↗](#); Westbrook et al., 2020 [↗](#)). We thus anticipated a positive correlation between choice and relative dwell time over the course of the

Figure 2. Example trial with the sampled value and accumulated value.

The sampled value (ΔSV ; red) and accumulated value (ΔAV ; black) are plotted for this example trial. For the first draw, ΔSV and ΔAV are identical. However, as the trial proceeds, the two signals diverge. In the model, a choice is made when the $|\Delta AV|$ reaches a pre-specified decision boundary.

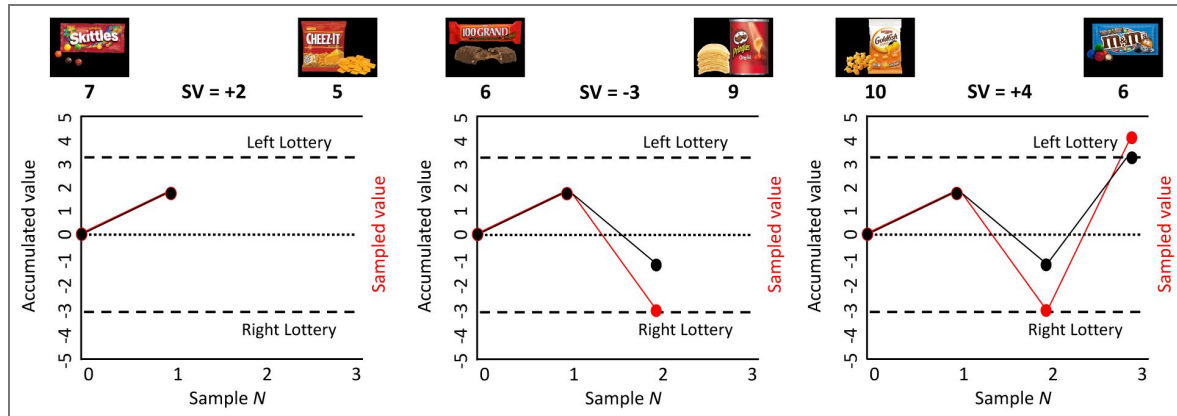
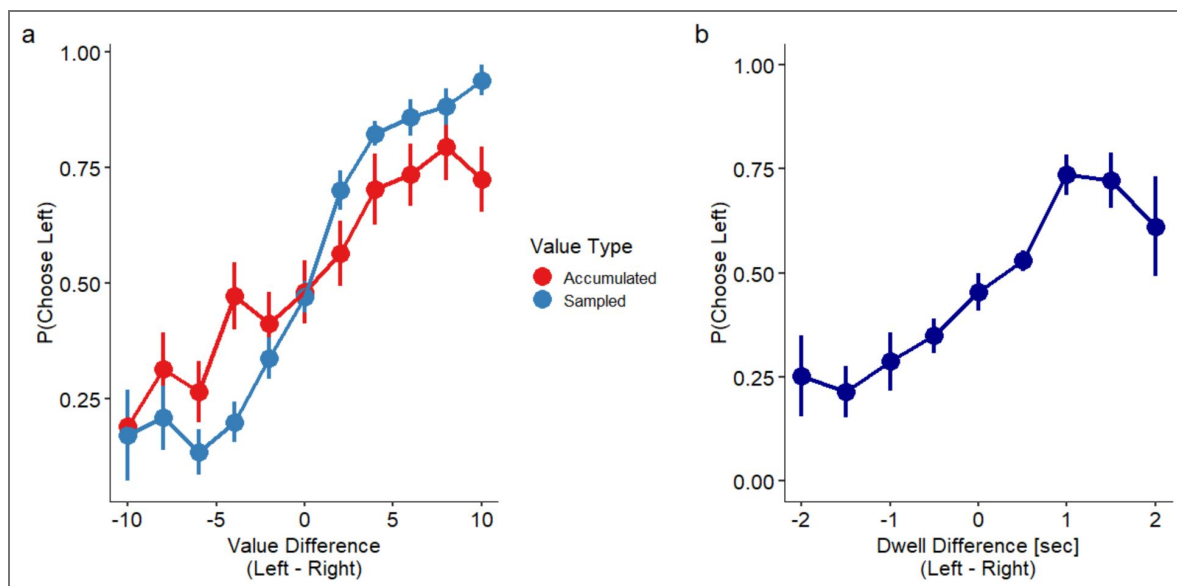


Figure 3. Choice data.

(a) The probability of choosing left based on ΔSV and ΔAV . As the value difference becomes greater in favor of one option, the probability of choosing that option increases, for both ΔSV and ΔAV . (b) The effect of gaze on choice. The longer that subjects looked at one lottery over the other, over the course of the whole trial, the more likely they were to choose that lottery.



whole trial. We added dwell proportion advantage (left – right dwell time divided by total dwell time) to the choice regression and observed a positive effect on choosing the left lottery ($\beta = 1.642$, $SE = 0.459$, $p < 0.001$).

To be certain that gaze-weighted evidence accumulated over the course of the whole trial, and not simply on the final sample of the trial, we excluded the final sample from each trial and re-ran the previous regression. All regressors were significant and positive (ΔSV : $\beta = 0.290$, $SE = 0.028$, $p < .001$; ΔAV : $\beta = 0.059$, $SE = 0.011$, $p < 0.001$; dwell proportion advantage: $\beta = 1.636$, $SE = 0.462$, $p < 0.001$). Thus, the influence of dwell time on choice occurred over the course of the entire decision, not simply on the final sample (Fig. 3B).

Neuroimaging results

Main analysis: Gaze-based contributions to sampled and accumulated evidence (GLM1)

Our general strategy for the fMRI data was to identify regions with BOLD activity correlating with the time series of $|\Delta SV|$ or $|\Delta AV|$. We primarily focused on the absolute value differences since we were looking for evidence in favor of making any choice, not specifically the left or right choice. This way we could identify the key components of the SSM choice process: the inputs and the integrators. We then tested whether these representations were modulated by gaze.

We tested the following hypotheses: (1) vmPFC and striatum contain input but not integrator representations; (2) the pre-supplementary motor area (pre-SMA), the intraparietal sulci (IPS), and dorsolateral PFC (dlPFC) contain integrator but not input representations; (3) the vmPFC and striatum input representations are modulated by gaze; (4) the pre-SMA, IPS, and dlPFC integrator representations are modulated by accumulated gaze.

To test all four hypotheses, we constructed a general linear model (GLM1) that included both non-gaze-weighted terms ($|\Delta SV|$ and lagged $|\Delta AV|$) and gaze-weighted terms ($|\Delta SV_{Gaze}|$ and lagged $|\Delta AV_{Gaze}|$). For hypotheses (1) and (2), the variables of interest were absolute sampled value difference ($|\Delta SV|$) and absolute accumulated value difference ($|\Delta AV|$) where $|\Delta AV| = |\sum_{t=1}^{T-1} \Delta SV_t|$, t is in units of draws (i.e., pairs of samples), and T is the current draw. Note again that we sum $|\Delta AV|$ to $T-1$ in order to exclude the current sample.

For hypotheses (3) and (4), the variables of interest were absolute gaze-weighted sampled values ($|\Delta SV_{Gaze}|$) and absolute gaze-weighted accumulated values ($|\Delta AV_{Gaze}|$). The sampled gaze-weighted values (ΔSV_{Gaze}) were derived by multiplying the proportion of left gaze time with the left value, and the proportion of right gaze time with the right value, within a sample. Accumulated gaze-weighted value differences (ΔAV_{Gaze}) were the sums of ΔSV_{Gaze} across samples. Again, $|\Delta AV_{Gaze}|$ did not include the current sample.

These regressors were interacted with boxcar functions covering each sample period (2 seconds). In addition to the regressors of interest, GLM1 contained a stick function for the button press onset, modulated by lagged $|\Delta AV|$, as a nuisance regressor event, as well as a boxcar function during the feedback screen, modulated by the value of the received item. We also added motion parameter time series to account for variation due to motion. We report results using FWE-corrected statistical significance of $p < 0.05$ and a cluster significance threshold of $p < 0.005$.

Hypothesis 1: vmPFC and striatum represent sampled but not accumulated value

We first investigated the vmPFC and striatum, regions that we hypothesized represent the inputs (i.e., SV). Looking specifically at BOLD activity in the vmPFC ROI defined in Bartra et al. (2013), we found a positive correlation with $|\Delta SV|$ (peak voxel $x = -14$, $y = 44$, $z = -12$; $t = 3.90$, $p = 0.019$) (Fig. 4A, 5A). However, we found no significant correlation with $|\Delta AV|$ in the vmPFC (Fig. 5A). The striatum showed no significant relationship with either $|\Delta SV|$ or $|\Delta AV|$ (Fig. 5A).

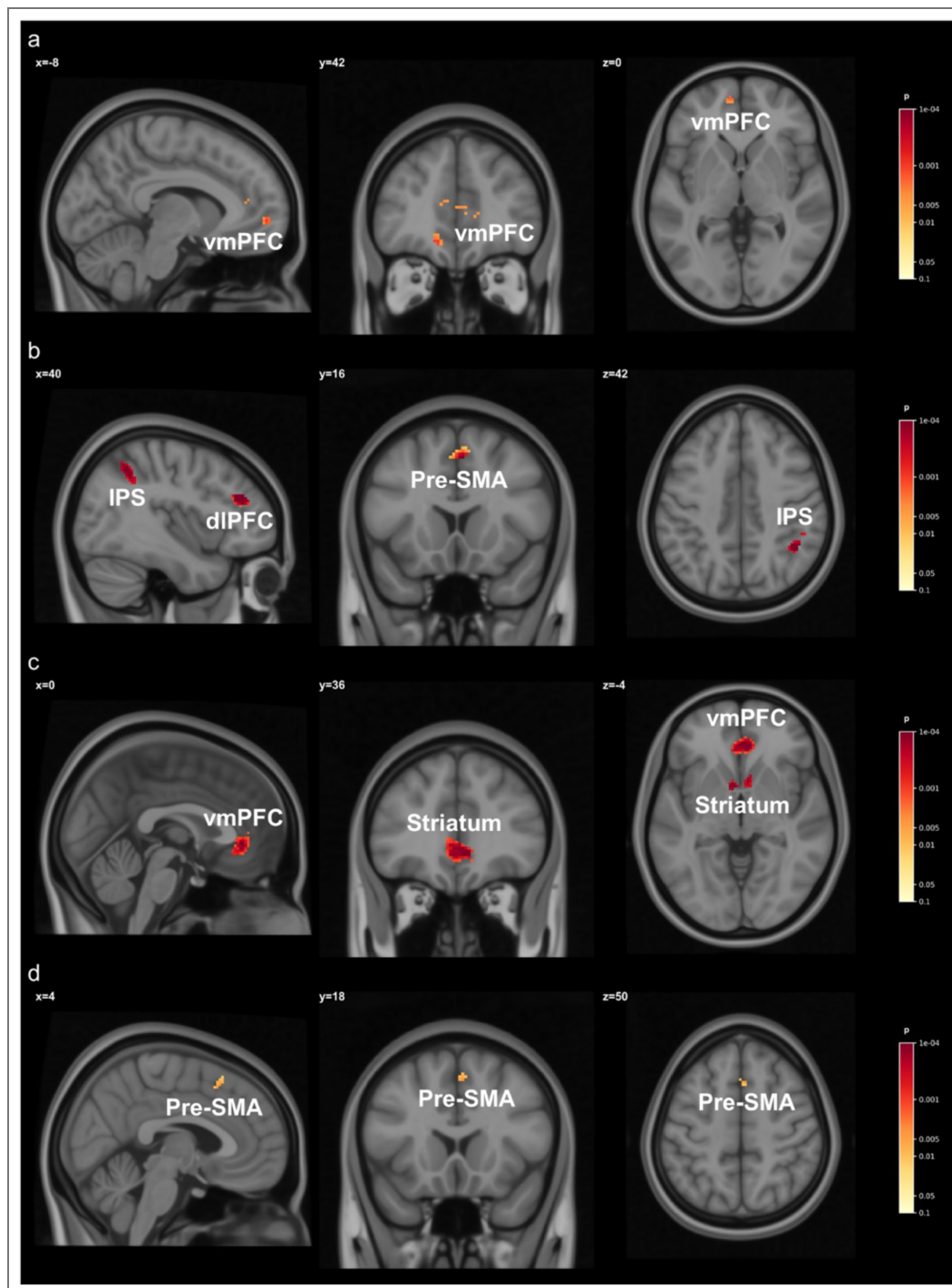


Figure 4. Regions responding to sampled value, accumulated value, and their gaze-weighted variants.

(a) vmPFC showed a significantly positive correlation with $|\Delta SV|$. (b) Pre-SMA, IPS, and dIPFC showed significantly positive correlations with lagged $|\Delta AV|$. (c) vmPFC and striatum showed significantly positive correlations with $|\Delta SV_{Gaze}|$, as did pre-SMA, IPS, and dIPFC (not pictured). (d) Pre-SMA showed a significantly positive correlation with lagged $|\Delta AV_{Gaze}|$.

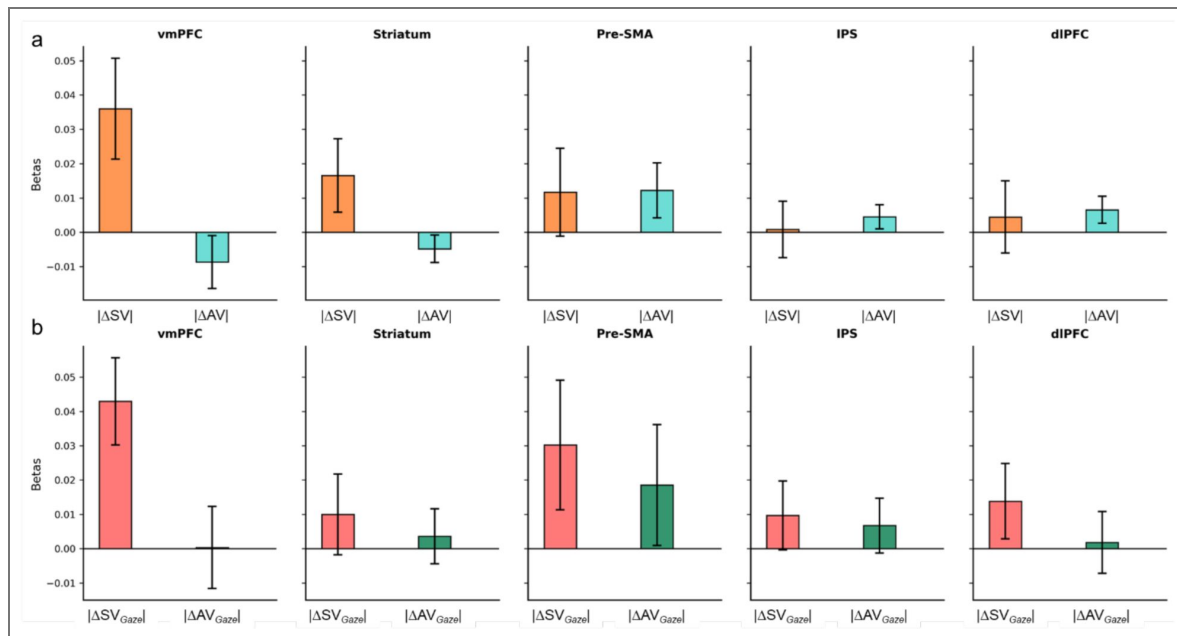


Figure 5. GLM1 beta plots from the vmPFC, striatum, pre-SMA, IPS, and dlPFC.

Displayed are regression coefficients from each region for (a) non-gaze-weighted signals: absolute sampled value difference ($|\Delta SV|$) and absolute lagged accumulated value difference ($|\Delta AV|$), and (b) gaze-weighted signals: gaze-weighted sampled value ($|\Delta SV_{Gaze}|$) and absolute lagged gaze-weighted accumulated value ($|\Delta AV_{Gaze}|$). vmPFC shows a pattern of BOLD activity that scales positively with both $|\Delta SV|$ and $|\Delta SV_{Gaze}|$, but does not respond to $|\Delta AV|$ or $|\Delta AV_{Gaze}|$. The striatum shows sensitivity only to $|\Delta SV_{Gaze}|$. In contrast, the pre-SMA, IPS, and dlPFC show strong positive correlations between BOLD activity and $|\Delta AV|$, but minimal relation with $|\Delta SV|$. The pre-SMA additionally shows sensitivity to $|\Delta AV_{Gaze}|$, indicating that accumulated value representations in this region are modulated by the history of gaze allocation. Note: Bar heights reflect mean betas averaged across all voxels within each ROI. Statistical significance was determined using permutation tests with FWE correction, which identify spatially localized, reliable effects within each ROI. For the striatum, while $|\Delta SV|$ shows a numerically larger overall mean, only $|\Delta SV_{Gaze}|$ produced a spatially consistent cluster of voxels surviving FWE correction (118 voxels, $p = 0.010$).

Hypothesis 2: pre-SMA, IPS, and dlPFC represent accumulated but not sampled value

We next investigated the pre-SMA, IPS, and dlPFC regions, which we hypothesized represent the integrated values (i.e., AV). In the pre-SMA, whose ROI we defined based on Hare et al. (2011) [\[4\]](#), we found a significant, positive relationship between BOLD activity and $|\Delta AV|$ (peak voxel: $x = 4$, $y = 14$, $z = 50$; $t = 4.75$, $p < 0.001$) (Fig. 4B [\[4\]](#), 5A [\[4\]](#)), but no relationship with $|\Delta SV|$ (Fig. 5A [\[4\]](#)). In the IPS, identified with the Harvard-Oxford Cortical Structural Atlas, we also saw a significant increase in activity as $|\Delta AV|$ increased (peak voxel: $x = 42$, $y = -50$, $z = 42$; $t = 5.16$, $p = 0.010$) (Fig. 4B [\[4\]](#), 5A [\[4\]](#)), but no relationship with $|\Delta SV|$ (Fig. 5A [\[4\]](#)). In the dlPFC, whose ROI we defined based on Hare et al. (2011) [\[4\]](#), we found the same pattern as the pre-SMA: there was a significantly positive relationship between BOLD activity and $|\Delta AV|$ (peak voxel: $x = 40$, $y = 34$, $z = 26$; $t = 6.83$, $p < 0.001$) (Fig. 4B [\[4\]](#), 5A [\[4\]](#)), and no relationship between BOLD activity and $|\Delta SV|$ (Fig. 5A [\[4\]](#)).

In summary, we found evidence that the vmPFC represents inputs but not integrators, while the pre-SMA, IPS, and dlPFC represent integrators but not inputs.

Hypothesis 3: vmPFC and striatum sampled value representations are modulated by gaze

Having identified input and integrator regions, we next asked whether the activity in these regions was affected by gaze. The aDDM (and other gaze-weighted SSMS) predict that gaze to one option should amplify that option's value relative to the other option (Krajbich et al., 2010 [\[4\]](#); Thomas et al., 2019 [\[4\]](#); Westbrook et al., 2020 [\[4\]](#)). Consider a simple model where an option's value is weighted by the proportion of time during which it is looked at. Imagine two trials with the same pair of values, 7 on the left and 3 on the right. In Trial A, the subject looks left 30% of the time and right 70% of the time. In Trial B, the subject looks left 70% of the time and right 30% of the time. In Trial A, the net input value ("drift rate") would be $|0.3 \cdot 7 - 0.7 \cdot 3| = 0$. In Trial B, the drift rate would be $|0.7 \cdot 7 - 0.3 \cdot 3| = 4$. In Trial A, the value advantage for the left option is canceled out by the gaze advantage for the right option. In Trial B, both the value and gaze advantage favor the left option, leading to strong evidence in favor of a left choice. In sum, there is stronger evidence when gaze difference is aligned with value difference. This should be true for both ΔSV and ΔAV , though ΔSV is only affected by gaze during the current draw, while ΔAV is affected by gaze over the entire trial.

In the vmPFC, we found a significant correlation with $|\Delta SV_{Gaze}|$ (peak voxel: $x = 4$, $y = 38$, $z = -4$; $t = 5.21$, $p = 0.004$) but no effect of $|\Delta AV_{Gaze}|$ (Fig. 4C [\[4\]](#), 5B [\[4\]](#)). Notably, the striatum also showed a significant correlation with $|\Delta SV_{Gaze}|$ (peak voxel: $x = 22$, $y = 20$, $z = -10$; $t = 5.10$, $p = 0.014$) (Fig. 4C [\[4\]](#), 5B [\[4\]](#)), but no corresponding effect for $|\Delta AV_{Gaze}|$ (Fig. 5B [\[4\]](#)). This result is particularly important because the striatum did not show sensitivity to non-gaze-weighted $|\Delta SV|$, suggesting that gaze modulation reveals striatal involvement in value processing that would otherwise be obscured.

Beyond these hypothesized relations, we also observed significant correlations between $|\Delta SV_{Gaze}|$ and activity in the IPS (peak voxel: $x = -14$, $y = -54$, $z = 48$; $t = 4.64$, $p = 0.026$), pre-SMA (peak voxel: $x = 6$, $y = 6$, $z = 58$; $t = 3.33$, $p = 0.031$), and dlPFC (peak voxel: $x = 46$, $y = 26$, $z = 14$; $t = 3.89$, $p = 0.034$) (Fig. 5B [\[4\]](#)). This is likely due to ΔSV contributing to the accumulated evidence, which would explain activity in the purported $|\Delta AV|$ regions. In other words, there is no reason why activity in accumulator regions should only reflect the lagged $|\Delta AV|$ variable and not the most recent evidence as well.

Hypothesis 4: pre-SMA, IPS, and dlPFC accumulated value representations are modulated by gaze

In the pre-SMA we found a significantly positive relationship with $|\Delta AV_{Gaze}|$ (peak voxel: $x = 4$, $y = 18$, $z = 50$; $t = 2.98$, $p = 0.032$) (Fig. 4D [\[4\]](#), 5B [\[4\]](#)). The IPS and dlPFC, on the other hand, did not show significant correlations with $|\Delta AV_{Gaze}|$ (Fig. 5B [\[4\]](#)).

Gaze allocation and value alignment (GLM2)

To more directly test whether accumulated evidence signals were modulated by accumulated gaze allocation throughout a trial, we conducted additional, exploratory analyses. Specifically, we ran a GLM that incorporated the following two terms: accumulated dwell advantage and $\Delta AV \times$ accumulated dwell advantage, in addition to ΔSV , the current gaze location, and $\Delta SV \times$ current gaze location.

We calculated accumulated dwell advantage as follows: For each sample t , accumulated dwell advantage is the cumulative difference in gaze allocation up to sample $t-1$, calculated as (total dwell left – total dwell right) / (total dwell left + total dwell right). This is a continuous measure from -1 (all previous gaze to right) to +1 (all previous gaze to left).

We also included the interaction between accumulated dwell advantage and ΔAV (i.e., signed accumulated evidence). This interaction term is positive when gaze is primarily to the left and left has more value or when gaze is primarily to the right and right has more value. This interaction term directly tests whether brain regions encoding accumulated evidence are modulated by the history of gaze allocation. This approach allows us to examine the interaction effect more explicitly as a separate regressor rather than having it embedded within the value calculation itself.

This GLM revealed a positive correlation between pre-SMA activity and the $\Delta AV \times$ accumulated dwell advantage term (peak voxel: $x = 8, y = 10, z = 58; t = 3.01, p = 0.026$). Surprisingly, the striatum also showed a correlation with this term (peak voxel: $x = -16, y = 10, z = -6; t = 4.07, p = 0.018$). Additionally, activity in the dlPFC was positively correlated with ΔSV (peak voxel: $x = -36, y = 34, z = 22; t = 3.96, p = 0.016$). No other ROIs showed significant relations.

This analysis provides additional evidence that the pre-SMA encodes accumulated value signals that are modulated by the history of gaze allocation.

Accounting for individual difference in attentional discounting (GLM3)

In our first set of analyses, we implicitly assumed complete discounting of non-fixated information, in contrast with previous studies that have generally found only partial discounting (Krajibich et al., 2010 [DOI](#); Sepulveda et al., 2020 [DOI](#); Smith & Krajibich, 2019 [DOI](#); Westbrook et al., 2020 [DOI](#)). To verify that our results are robust to inter-subject variability in attentional discounting, we estimated subject-level attentional discounting parameters and then re-estimated our original GLM with new, recalculated gaze-weighted value regressors.

Following Smith, Krajibich, and Webb (2019) [DOI](#), for each individual sample within a trial, we computed:

$$\beta = (G_{\text{Left}} \times V_{\text{Left}}) - (G_{\text{Right}} \times V_{\text{Right}})$$

$$\gamma = (G_{\text{Right}} \times V_{\text{Left}}) - (G_{\text{Left}} \times V_{\text{Right}}),$$

where G_{Left} and G_{Right} represent the proportion of time spent gazing left versus right within that specific sample, and V_{Left} and V_{Right} are the instantaneous values of the left and right options. We then averaged these sample-level β and γ values across all samples within each trial to obtain trial-level regressors. We then ran a mixed-effects logistic regression predicting choice (left vs. right) as a function of β and γ and then calculated subject-specific values of $\theta = \gamma/\beta$. Across our sample ($N=20$), we found mean $\theta = 0.77$ (SD = 0.21, range = 0.55–1.25).

Next, for the GLM, we computed θ -weighted sampled-value ($|\Delta SV_{\theta}|$) as:

$$\Delta SV_{\theta} = \left(G_{\text{Left}} \times (V_{\text{Left}} - \theta V_{\text{Right}}) \right) - \left(G_{\text{Right}} \times (V_{\text{Right}} - \theta V_{\text{Left}}) \right)$$

Similar to the original GLM, we computed an accumulated value signal, $|\Delta AV_{\theta}|$, based on the lagged sum of previous samples' $|\Delta SV_{\theta}|$.

We found significant positive effects of $|\Delta SV_{\theta}|$ in the vmPFC (peak voxel: $x = -14, y = 44, z = -12; t = 3.57, p = 0.027$) and IPS (peak voxel: $x = 30, y = -28, z = 40; t = 4.58, p = 0.020$), but in no other ROI.

In contrast, we found significant positive relationships between $|\Delta AV_\theta|$ and activity in the pre-SMA (peak voxel: $x = 0, y = 22, z = 52$; $t = 4.68, p = 0.001$), dlPFC (peak voxel: $x = 40, y = 32, z = 26$; $t = 4.32, p = 0.004$), and IPS (peak voxel: $x = 44, y = -48, z = 42$; $t = 6.26, p < 0.0001$). Notably, we also observed a significant relationship between $|\Delta AV_\theta|$ and activity in the vmPFC ($x = 8, y = 38, z = 18$; $t = 3.89, p = 0.041$). No other significant contrasts emerged.

In summary, these analyses provide additional evidence that the vmPFC encodes gaze-weighted sampled value signals and the pre-SMA encodes gaze-weighted accumulated value signals, though other correlations also emerged.

Modeling a non-uniform temporal weighting function (GLM4)

While sequential sampling models like the DDM assume equal weighting of information during evidence accumulation, other models allow for information sampled at different time points to differentially impact choice. For example, information that arrives early (i.e., primacy) or late (i.e., recency) can preferentially influence decision-making (Usher & McClelland, 2001). This could especially be the case in our task where information has to be integrated over a long period of time.

To account for temporal biases during evidence accumulation, we fit participant data with a model that incorporates primacy and recency biases into the sequential sampling process (Galdo et al., 2022; Pooley et al., 2011). Within a trial t , the weight of sample i is determined by the following temporal weighting function:

$$w_t(i) = [1 - (1 - \varepsilon_p^i)(1 - \varepsilon_r^{N_t - i + 1})](1 - \eta) + \eta, \quad (1)$$

where ε_p and ε_r are weights on primacy and recency, respectively, η is a lower bound on the weight of any sample, and N_t is the number of samples on trial t .

In the context of the current experiment, we assume that decision-makers accumulate evidence (ΔAV) based on the sum of sampled evidence (ΔSV) weighted by the temporal weighting function:

$$\Delta AV_{non-uniform,t} = \sum_{N_t}^i w_t(i) \Delta SV_t(i). \quad (2)$$

At the time of choice, the decision-maker chooses according to the logit function:

$$P_t(\text{Left Lottery}) = \frac{1}{1 + e^{-\beta \Delta AV_{non-uniform,t}}} \quad (3)$$

with inverse-temperature parameter β governing the how strongly $\Delta AV_{non-uniform,t}$ governs the selection of the higher value lottery.

For models incorporating visual attention, the only modification is to use ΔSV_{Gaze} instead of ΔSV .

Behavioral model results

We observed a substantial recency bias, with all participants showing $\varepsilon_r > \varepsilon_p$ when using ΔSV and all but one participant showing $\varepsilon_r > \varepsilon_p$ when using ΔSV_{Gaze} (Fig. 6) as inputs. Independent samples t-tests (with the Welch approximation to degrees of freedom) confirmed that participants' ε_r parameters were significantly larger than their ε_p parameters, for both ΔSV ($t(36) = 12.54, p = 10^{-14}$) and ΔSV_{Gaze} ($t(20) = 7.25, p = 10^{-6}$) inputs. Goodness-of-fit measures based on the Bayesian Information Criterion (BIC) also favored a non-uniform temporal weighting function (No gaze: BIC = 1262; Gaze: BIC = 1593) over a uniform temporal weighting function (No gaze: BIC = 1307; Gaze: BIC = 1658). However, it is worth noting that the recency bias is surely over-estimated based on our parameter-recovery exercise (Methods) and the bias introduced by allowing participants to choose when to stop collecting evidence (Discussion).

fMRI model results

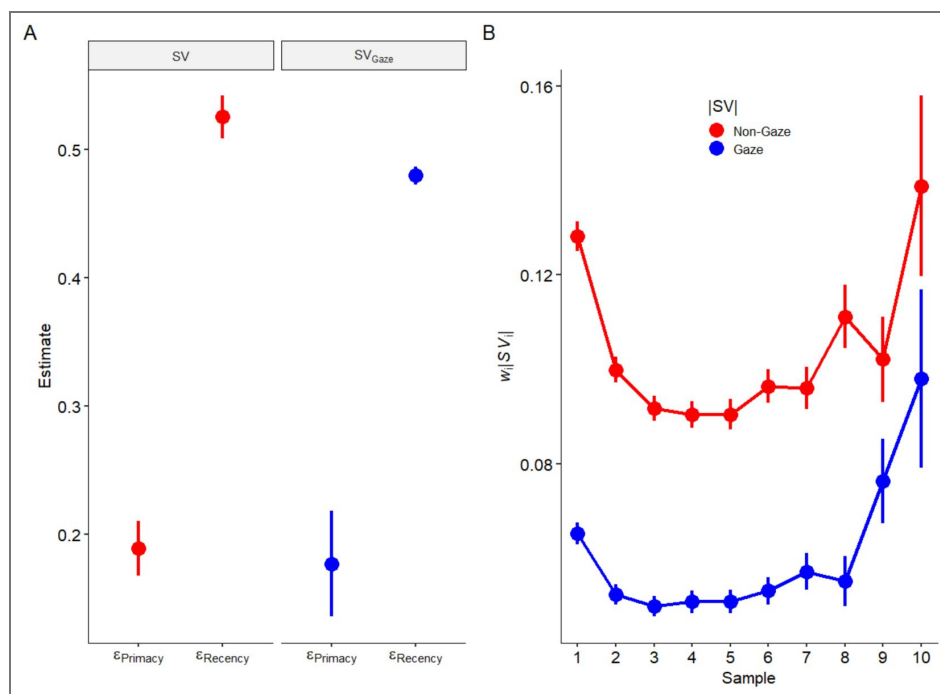


Figure 6. Non-uniform temporal weighting.

In both gaze-weighted and non-gaze-weighted models, participants showed stronger recency than primacy effects, both in terms of **(A)** the model parameters, and **(B)** the resulting temporal weighting functions averaged across all trials. Error bars are standard errors clustered by participant.

To identify brain regions that encoded $|\Delta AV_{non-uniform}|$, we tested whether activation parametrically varied as a function of these new values using updated versions of GLM1 in the previously defined ROIs. Within each ROI, we used a FWE corrected threshold of $p < 0.05$ and cluster-forming threshold of $p < 0.005$ with 5000 permutations.

We again found significant correlations between $|\Delta AV_{non-uniform}|$ and BOLD activity in the pre-SMA (peak voxel $x = 0, y = 22, z = 52$; $t = 4.68, p = 0.001$), IPS (peak voxel $x = 44, y = -48, z = 42$; $t = 6.23, p < 0.001$), and dlPFC (peak voxel $x = 40, y = 32, z = 26$; $t = 4.32, p = 0.052$). Additionally, we found a correlation in the vmPFC (peak voxel $x = 8, y = 38, z = 18$; $t = 3.89, p = 0.037$).

For gaze-weighted $|\Delta AV_{non-uniform}|$, we found significant correlations with BOLD activity in the pre-SMA (peak voxel $x = 4, y = 18, z = 50$; $t = 2.80, p = 0.047$) and dlPFC (peak voxel $x = 38, y = 30, z = 26$; $t = 3.86, p = 0.012$).

Discussion

In this article we presented results from a simultaneous eye-tracking and fMRI study of value-based decision-making, using an expanded-judgment task where subjects sampled from, and then chose between food lotteries. We found that the vmPFC, and to a lesser extent the striatum, represent sampled input values, and the pre-SMA, IPS, and dlPFC appear to compute accumulated values. We found that sampled value signals in vmPFC and striatum are modulated by gaze allocation (Lim et al., 2011 [↗](#); McGinty et al., 2016 [↗](#)), and that this gaze modulation extends to accumulated value signals in the pre-SMA.

These results provide novel evidence for the neural mechanisms underlying the SSM process, as exemplified by the DDM, which appears to govern many types of decisions (Busemeyer, 1985 [↗](#); Ratcliff, 1978 [↗](#)). The gaze modulation of the accumulated value signals in the pre-SMA provides critical evidence that this region indeed represents accumulated evidence, as opposed to unchosen values (Boorman et al., 2011 [↗](#); Kolling et al., 2016 [↗](#); Wittmann et al., 2016 [↗](#)), decision conflict (Frömer et al., 2019 [↗](#); Kaanders et al., 2021 [↗](#); Shenhav et al., 2014 [↗](#), 2016 [↗](#); Vassena et al., 2020 [↗](#)), or time on task (Grinband et al., 2011 [↗](#); Holroyd et al., 2018 [↗](#); Mumford et al., 2024 [↗](#)). While accumulated evidence is typically correlated with these other measures, we were able to dissociate them by taking advantage of the fact that accumulated evidence, but not the other measures, are modulated by gaze location. Value is amplified by gaze (Smith & Krajbich, 2019 [↗](#)), leading to stronger value signals in the brain when the decision maker is looking at the higher value option. This is what we observed in the vmPFC and striatum for input values, and in the pre-SMA for accumulated values – consistent with an SSM account.

Conflict-based models make qualitatively different predictions. Regions implementing conflict monitoring should show increased activity when options are similar in value, regardless of time. The conflict account predicts that BOLD activity should scale with inverse value difference: smaller $|\Delta V| \rightarrow$ higher conflict $\rightarrow$ higher BOLD (Shenhav et al., 2014 [↗](#), 2016 [↗](#)). In simple choice tasks, high conflict and high accumulated value are both associated with long response times (Pisauro et al., 2017 [↗](#)), leading to ambiguity about how to interpret purported neural correlates of accumulated value. In our task we avoided this ambiguity by analyzing the effect of accumulated value at each point in time, not just at the moment of decision. With this approach, conflict should be inversely correlated with accumulated value (as higher accumulated evidence indicates less similarity between options). Moreover, the conflict account makes no predictions about how BOLD activity should be modulated by gaze allocation for a given set of option values.

A more serious concern is the potential confound with time-on-task BOLD activity. Accumulated value inevitably increases with time within a trial, leading to a correlation between the two variables (Grinband et al., 2011 [↗](#); Holroyd et al., 2018 [↗](#); Mumford et al., 2024 [↗](#)). This is where the gaze data were particularly important. Time-on-task regions should show no relation with gaze allocation patterns. After accounting for non-gaze-weighted accumulated value, only accumulator regions, and not time-on-task regions, should show a relationship with gaze-weighted

accumulated value. The results of our analyses provide exactly such evidence: pre-SMA activity was positively correlated with gaze-weighted accumulated value, even when accounting for previous gaze history and individual differences in attention discounting.

These findings were made possible by considering the role that visual attention plays in the decision process. While SSMs capture choice behavior and RTs extremely well, most do not consider the effects of attention. Attention is thought to shift over the course of the decision, amplifying the attended inputs and/or inhibiting the non-attended inputs (Diederich, 1997; Johnson & Busemeyer, 2005; Roe et al., 2001). These shifts in attention are reflected in eye-movements (Hoffman & Subramaniam, 1995), which affect choice outcomes (Fiedler et al., 2013; Fiedler & Glöckner, 2015; Folke et al., 2016; Glaholt & Reingold, 2009; Gwinn et al., 2019; Janiszewski et al., 2013; Jiang et al., 2016; Kim et al., 2012; Konovalov & Krajbich, 2016; Lopez-Persem et al., 2016; Orquin & Mueller Loose, 2013; Pärnamets et al., 2015; Polonio et al., 2015; Russo & Leclerc, 1994; Sheng et al., 2020; Shi et al., 2012; Smith & Krajbich, 2018; Stewart et al., 2016; Tavares et al., 2017; Towal et al., 2013; Vaidya & Fellows, 2015; Vanunu et al., 2021; Venkatraman et al., 2014; J. T. Wang et al., 2010), and their effect on the choice process is captured by the attentional drift diffusion model (aDDM) and other related SSMs (Ashby et al., 2016; Fisher, 2021; Glickman et al., 2019; Jang et al., 2021; Krajbich et al., 2010; Li & Ma, 2021; Smith & Krajbich, 2019; Teoh et al., 2020; Westbrook et al., 2020; Yang & Krajbich, 2023; Zilker & Pachur, 2023).

Neural implementations of SSMs generally require at least two sets of neurons, one set to represent the current information from the stimuli and a second set to integrate that information over time. In the current study, the information being used to make decisions was subjective value (i.e., utility). A large body of work has implicated the vmPFC and striatum in representing value (Bartra et al., 2013; Boorman et al., 2009; Chib et al., 2009; Levy & Glimcher, 2011). Interestingly, our results confirm that these two regions do represent value information, but primarily just for the presented stimuli. The integrated value information, which is what ultimately determines the decision, is instead encoded in the pre-SMA, IPS, and dlPFC, regions that have received less attention in the literature.

By using eye-tracking, our study extends previous work connecting computational models to fMRI data. Our results align with Hare et al.'s (2011) proposed neural model in which the vmPFC provides inputs to the pre-SMA and IPS. While the Hare model was based on dynamic causal modeling results, our task provides a more direct test of the proposed neural network. Additionally, our eye-tracking data allows us to identify additional features of the network. First, we identify the striatum as an input region, a result that only appears in analyses that account for gaze-weighted value (Lim et al., 2011). Second, we find that the pre-SMA is sensitive to gaze-weighted accumulated value, while the IPS is not. The reason for this distinction between the pre-SMA and IPS is unclear, but it does suggest that the pre-SMA is more likely to be the final decision-making region, consistent with some recent studies (Juechems et al., 2017; PISAURO et al., 2017; Rodriguez et al., 2015; Rouault et al., 2019; Aquino et al., 2023). Of course, the recruitment of the pre-SMA may be because our subjects made their decisions with a button press, which is supported by the correlations between accumulated value and motor cortex activity. Had our study required eye-movements to indicate a choice, we may very well have observed integrator activity in other regions such as the frontal eye fields or posterior parietal cortex (O'Connell et al., 2018).

A major advantage of this study is its use of a task designed to slow down the decision process and force sequential integration of information. Such expanded judgment tasks have been used to study SSM assumptions in perceptual decision-making, more recently in combination with neural recordings, but mostly with electrophysiology in rats and monkeys (Brunton et al., 2013; Cisek et al., 2009; Gluth et al., 2012; Tsetsos et al., 2012; T. Yang & Shadlen, 2007).

On the other hand, one concern with longer decision times is that decision-makers might either under-weight (i.e., forget), or put too much weight on, early information. Our analysis of subjects' temporal weighting functions did reveal a primacy effect, where the first sample in each trial was overweighted, as well as a recency effect, where the last samples in each trial were also

overweighted. Nonetheless, using a temporal weighting function with these primacy and recency effects did not substantially change the conclusions from our fMRI analysis. While these analyses did reveal weak support for accumulator dynamics in the vmPFC, these results should be interpreted with caution because adding recency effects into the accumulated-value signal increased the correlation between $|\Delta SV|$ and $|\Delta AV|$ from 0.12 to 0.21. Moreover, the recency effect is surely overestimated due to well-known statistical artifacts (Mullett & Stewart, 2016 [↗](#)). In short, because decision-makers tend to terminate the decision process after strong pieces of evidence, and because a strong piece of evidence will tend to have a large noise component, information that appears at the end of a trial will appear to have a stronger influence on choices than it should.

Another potential limitation of our study was the number of participants. While our sample size of 20 subjects is modest by current neuroimaging standards, the within-subject statistical power from our extended decision paradigm (~380 observations per subject), combined with hypothesis-driven ROI analyses and multiple comparisons correction, provides confidence in our core findings. Nevertheless, replication with larger samples would be valuable, particularly for more fully characterizing null effects and marginal findings.

Taken together, we separated sampled input values from the overall decision value, or accumulated value, and found a network of brain regions that are involved in an addm-like choice process. This process involves passing sampled input values to an integrator which responds to not only the values themselves, but also to the gaze-modulated values. These results indicate that gaze effects on value representations are not epiphenomenal; rather, they reflect how gaze is incorporated into the decision process, affecting how we perceive the value of each option over the course of the entire decision.

Materials and Methods

Experimental Design and Statistical Analyses

Subjects

Twenty-eight undergraduate students at The Ohio State University participated in this study. Due to time constraints, 4 subjects were unable to finish all 3 runs of the scan. One subject was excluded for not choosing in line with their ratings. This left 23 subjects in the analysis (14 men, 9 women, average age: 22.61). Another 3 subjects could not be calibrated on the eye-tracker and so are discarded from any analyses involving eye tracking (leaving 13 men, 7 women, average age: 22.9). All subjects were right-handed, had normal or corrected-to-normal vision, and no history of neurological disorders. This study was approved by The Ohio State Biomedical Sciences IRB.

Stimuli and Tasks

Rating Task

Outside of the scanner, subjects rated 148 food items using a continuous rating scale from -10 (extreme dislike) to 10 (extreme like), with 0 being indifference towards an item. To choose their preferred rating, subjects moved the mouse across the rating scale and then clicked the left mouse button when the cursor was at their desired value for the item. We used an incentivization procedure for these ratings. There was a 50% chance that the rating task would be used to determine the subject's reward. In such cases, the computer would randomly select two foods and the subject would receive the one with the higher rating. If both items were rated negatively, the subject would not receive any food. Negative values (-10 to 0) were excluded from the choice task except for two subjects who did not have enough positively valued items.

Choice Task

Once in the MRI scanner, subjects chose between pairs of lotteries. Lotteries were constructed by creating 1000 potential lotteries of randomly selected items evenly split between having 3, 4, 5, or 6 items. Each item in a lottery was then assigned a probability of being drawn, with probabilities

summing up to 1 in each lottery. For each of these lotteries we calculated its expected utility by multiplying the subjective value (i.e., rating) of each item (V_i) in the lottery by its associated probability (P_i) of being drawn, and summing the results

$$EU = \sum_{i=1}^N P_i * V_i \quad (4)$$

where N is the total number of items in that lottery. Subjects were not told these probabilities, nor were they told which items were in each lottery – they had to learn this every trial.

We tracked subjects' eye movements in the scanner using an Eyelink 1000 plus (SR Research) set at 500 Hz. Eye position was monitored with the camera and infrared source reflected in the mirror attached to the head coil. The eye tracker was calibrated at the beginning of the session.

Food items were sampled from the lotteries and presented on the screen one pair at a time. Each draw was presented for 2 seconds, followed by a fixation cross for 2-6 seconds. This process repeated until the subject made a choice. Once a subject was ready to choose a lottery, they used the index finger of their left (right) hand to press a button corresponding to the left (right) lottery. They were then presented with a random item from the lottery they had chosen (on the same side of the screen as the chosen basket) for 2 seconds, indicating the food they would receive from this trial, should it be randomly selected at the end of the study (Fig. 1 [↗](#)).

We constructed each trial's sequence of items pseudo-randomly to minimize the correlation between the sampled value signal ($|\Delta SV|$; i.e., the absolute difference in sampled input values) and the accumulated value signal ($|\Delta AV|$; i.e., the absolute difference in accumulated values). For the first draw in each trial, sampled and accumulated value signals are equal. On subsequent draws, the $|\Delta SV|$ diverges from the $|\Delta AV|$ signal, yielding two distinct time courses to look for in the fMRI data (Fig. 2 [↗](#)). Across subjects, $|\Delta SV|$ and $|\Delta AV|$ had an average correlation of 0.31 (SD = 0.10, min = 0.08, max = 0.45), while $|\Delta SV|$ and lagged $|\Delta AV|$ (i.e., the variables in our GLMs) had an average correlation of 0.07 (SD = 0.08, min = -0.09, max = 0.17). See Figure 7 [↗](#) for the full correlation matrix of value signal regressors calculated across subjects and samples.

The task structure was designed to incentivize subjects to average 7 samples per trial. They had 45 minutes to make 60 choices, and any trials that they did not complete by the end of the experiment were made for them randomly by the computer. Any trials they completed beyond 60 were simply added to the pool of potentially rewarded trials. At the end of the study, there was a 50% chance that one of the choice trials would be randomly selected for payment (otherwise the rating task was selected for payment), in which case the subject received the corresponding food from that trial.

Outside of the scanner, subjects first completed a 5-minute practice section where they chose between baskets made up of cars. After each trial, they were given feedback on how many samples they took and were reminded that the goal was to take 7 samples on average. These choices were not incentivized.

Temporal-weighting-function model fits

We estimated model parameters using an iterative maximum a posteriori (MAP) approach (Huys et al., 2011 [↗](#); Wittmann et al., 2020 [↗](#)). This method improves upon maximum likelihood estimation (MLE) by simultaneously estimating parameters at both the subject- and group-level. This hierarchical procedure constrains subject-level parameters and reduces the influence of outlier data.

Group-level parameters were initialized with uninformative Gaussian priors with mean of 0.1 and variance of 100. For all models, η was held constant at 1. During the expectation step, we estimated model parameters (ε_p , ε_r , β) for each participant using MLE and calculated the log-likelihood of their choices given the model parameters. During the maximization step, we calculated the maximum posterior probability based on the observed choices and prior group-level parameters, and then updated the group-level parameters to generate posterior parameter estimates. These posterior parameter estimates were then used as the priors in subsequent steps in this procedure. We iteratively repeated the expectation and maximization steps until convergence of the posterior

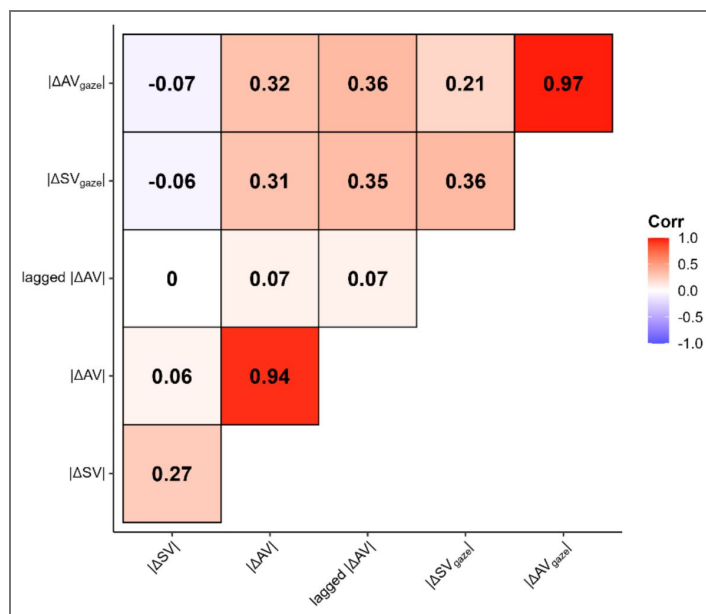


Figure 7. Sample-level correlations between value signal regressors.

Pearson correlations computed across samples for different quantifications of sampled value (ΔSV) and accumulated value (ΔAV). Values represent correlations between absolute magnitudes of each measure. $|\Delta SV|$ = absolute sampled value difference; $|\Delta AV|$ = absolute accumulated value difference; lagged $|\Delta AV|$ = absolute accumulated value difference up to the previous sample; $|\Delta SV_{gaze}|$ = gaze-weighted sampled value; $|\Delta AV_{gaze}|$ = gaze-weighted accumulated value; lagged $|\Delta AV_{gaze}|$ = gaze-weighted accumulated value from the previous sample.

likelihood summed over group-level parameters exceeded a change of less than 0.0001 from the previous iteration (for a maximum of 800 iterations). During this procedure, bounded free parameters were transformed from Gaussian space to native model space using link functions (e.g., sigmoidal function for ε_p , ε_r) to ensure accurate estimation near the bounds.

We assessed parameter recovery of this model on simulated choices using best-fitting model parameters. We then refit the simulated choices using the same MAP process described above. We found strong Pearson's r correlations between the generated and estimated parameter values ($r > .8$; Fig. 8 [↗](#)), though both parameters were systematically over-estimated.

MRI Data Acquisition

MRI scanning was carried out at the OSU Center for Cognitive and Behavioral Brain Imaging. We used a 3T Siemens Magnetom Prisma scanner with a 32-channel head array coil to collect the neural data. Functional data were acquired with a T2*-weighted gradient-echo sequence (48 slices, interleaved, with a field of view of 1554×1554, with an in-plane resolution of 3 mm isotropic and a 3mm slice gap, TR = 2600 ms, TE = 28 ms, 80° flip angle). Slices were oriented such that the anterior side of the acquisition was raised dorsally by 30 degrees compared to the line formed by joining the anterior commissure to the posterior commissure. A high-resolution MPAGE anatomical scan (256 slices, field of view 224×256, with an in-plane resolution of 1 mm and no slice gap, TR = 1900 ms, TE = 4.44 ms, 12° flip angle) was also acquired for each participant. Each participant was scanned in one 1.5-hour session, which included the three experimental runs (15 minutes each) and the high-resolution MPAGE anatomical scan. Additionally, a resting state scan (5 minutes) and a DTI scan were acquired, but these data are not presented here. Stimuli were presented using Psychtoolbox (Brainard, 1997 [↗](#); Kleiner et al., 2007 [↗](#); Pelli, 1997 [↗](#)) for MATLAB (MATLAB, 2016 [↗](#)), and displayed with a DLP projector onto a screen mounted in the rear of the scanner bore.

MRI Preprocessing and analyses

Statistical parametric mapping (SPM12, Update Rev. Nr. 6905; Functional Imaging Laboratory, University College London) was used to carry out the preprocessing of fMRI data. First, we corrected for the different slice times per echo planar image (EPI) across the total volume (using the bottom slice as a reference) and then realigned each volume in a run to the mean EPI volume from that run. Next, the anatomical scan was coregistered with the MNI average of 152 brains template, and the mean EPI per run was used to coregister all functional scans to this coregistered anatomical scan. In order to warp the EPIs to MNI space, SPM12's *normalise* function was applied to the coregistered anatomical scan and the resulting warping parameters were applied to the coregistered EPIs. The resulting images were smoothed using an isometric Gaussian kernel (8 mm full width at half maximum). First level GLMs were run using SPM on each subject individually, including contrasts of interest. Runs were combined using *fslmerge*. We then used the Statistical non-Parametric Mapping toolbox (e.g., SnPM013.1.09) function to run second-level, non-parametric significance tests with threshold-free clustering and family wise error (FWE) correction to find significant clusters for the described effects (Nichols & Holmes, 2002 [↗](#)). We report the results of 5000 permutations using FWE-corrected statistical significance of $p < 0.05$ and a cluster significance threshold of $p < 0.005$.

We used a canonical hemodynamic response function (HRF) without time derivatives. We modeled the noise as an AR(1) process. We additionally used a high-pass filter set to 128 s. We also used global signal normalization with a value of 0.8. We used no corrections for susceptibility distortions. All general linear models (GLM) included variants of $|\Delta SV|$ and lagged $|\Delta AV|$, either gaze weighted or not, interacted with boxcar functions covering each sample period (2 seconds) – see details below. In addition to the regressors of interest, each GLM contained the following nuisance terms: a stick function for trial number, a stick function for the button press onset modulated by lagged $|\Delta AV_{Gaze}|$, and a boxcar function during the feedback screen, modulated by the value of the received item. We also added motion parameter time series to account for variation due to motion.

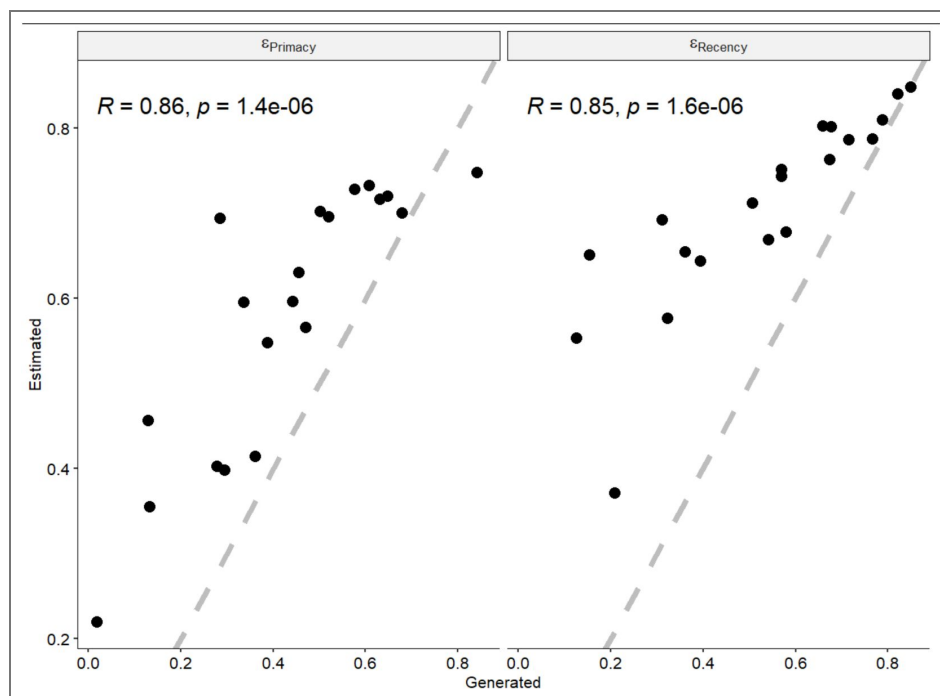


Figure 8. Correlation plots illustrating parameter recovery.

In our primary analyses (GLM1), in addition to the nuisance terms, we included the following regressors: $|\Delta SV|$, lagged $|\Delta AV|$, $|\Delta SV_{Gaze}|$ and lagged $|\Delta AV_{Gaze}|$.

In our analyses of gaze allocation history (GLM2), in addition to the nuisance terms, we included the following regressors: ΔSV , lagged ΔAV , current gaze location, accumulated dwell advantage, $\Delta SV \times$ current gaze location, and lagged $\Delta AV \times$ accumulated dwell advantage.

In our analysis incorporating individual differences in attentional discount (GLM3), in addition to the nuisance terms, we included the following regressors: θ -weighted $|\Delta SV|$ and lagged θ -weighted $|\Delta AV|$.

In our analysis incorporating non-uniform temporal weighting of accumulated evidence (GLM4), in addition to the nuisance terms, we included the following regressors: $|\Delta SV|$, lagged $|\Delta AV_{non-uniform}|$, $|\Delta SV_{Gaze}|$ and lagged $|\Delta AV_{non-uniform, Gaze}|$.

None of the regressors in the models were orthogonalized.

Region of interest specifications

ROIs were based upon previously published brain atlas parcellations and relevant literature. We used the Harvard-Oxford atlas for the intraparietal sulcus (IPS) and striatum (Desikan et al., 2006). The dorsolateral prefrontal cortex (dlPFC) and pre-supplementary motor area (pre-SMA) were defined based on (Hare et al., 2011). The ventromedial prefrontal cortex (vmPFC) was defined in (Bartra et al., 2013).

Data availability

Experiment and analysis code as well as Behavioral and eye-tracking data are available on the Open Science Framework: <https://osf.io/eyxvb/>. fMRI data is available on OpenNeuro: <https://doi.org/10.18112/openneuro.ds007305.v1.0.0>

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References

- Aquino T.G., Cockburn J., Mamelak A.N., Rutishauser U., O'Doherty J.P (2023) Neurons in human pre-supplementary motor area encode key computations for value-based choice. *Nature Human Behaviour* **7**:970-985 <https://doi.org/10.1038/s41562-023-01548-2> | PubMed
- Ashby N. J. S., Jekel M., Dickert S., Glöckner A (2016) Finding the right fit: A comparison of process assumptions underlying popular drift-diffusion models. *Journal of Experimental Psychology: Learning, Memory, and Cognition* **42**:1982-1993 <https://doi.org/10.1037/xlm0000279> | PubMed
- Bartra O., McGuire J. T., Kable J. W (2013) The valuation system: A coordinate-based meta-analysis of BOLD fMRI experiments examining neural correlates of subjective value. *NeuroImage* **76**:412-427 <https://doi.org/10.1016/j.neuroimage.2013.02.063> | PubMed
- Bhatnagar R., Orquin J. L (2022) A meta-analysis on the effect of visual attention on choice. *Journal of Experimental Psychology: General* **151**:2265 <https://doi.org/10.1037/xge0001204> | PubMed
- Bogacz R., Brown E., Moehlis J., Holmes P., Cohen J. D (2006) The physics of optimal decision making: A formal analysis of models of performance in two-alternative forced-choice tasks. *Psychological Review* **113**:700 <https://doi.org/10.1037/0033-295X.113.4.700> | PubMed

- Boorman E. D., Behrens T. E. J., Woolrich M. W., Rushworth M. F. S (2009) How Green Is the Grass on the Other Side? Frontopolar Cortex and the Evidence in Favor of Alternative Courses of Action. *Neuron* **62**:733-743 <https://doi.org/10.1016/j.neuron.2009.05.014> | PubMed
- Boorman E. D., Behrens T. E., Rushworth M. F (2011) Counterfactual Choice and Learning in a Neural Network Centered on Human Lateral Frontopolar Cortex. *PLOS Biology* **9**:e1001093 <https://doi.org/10.1371/journal.pbio.1001093> | PubMed
- Brainard D. H (1997) The Psychophysics Toolbox. *Spatial Vision* **10**:433-436 <https://doi.org/10.1163/156856897X00357> | PubMed
- Brunton B. W., Botvinick M. M., Brody C. D (2013) Rats and humans can optimally accumulate evidence for decision-making. *Science* **340**:95-98 <https://doi.org/10.1126/science.1233912> | PubMed
- Busemeyer J. R (1985) Decision making under uncertainty: A comparison of simple scalability, fixed-sample, and sequential-sampling models. *Journal of Experimental Psychology: Learning, Memory, and Cognition* **11**:538-564 <https://doi.org/10.1037/0278-7393.11.3.538>
- Busemeyer J. R., Townsend J. T (1993) Decision field theory: A dynamic-cognitive approach to decision making in an uncertain environment. *Psychological Review* **100**:432-459 <https://doi.org/10.1037/0033-295X.100.3.432> | PubMed
- Callaway F., Rangel A., Griffiths T. L (2021) Fixation patterns in simple choice reflect optimal information sampling. *PLOS Computational Biology* **17**:e1008863 <https://doi.org/10.1371/journal.pcbi.1008863> | PubMed
- Cavanagh J. F., Wiecki T. V., Kochar A., Frank M. J (2014) Eye tracking and pupillometry are indicators of dissociable latent decision processes. *Journal of Experimental Psychology: General* **143**:1476-1488 <https://doi.org/10.1037/a0035813> | PubMed
- Chib V. S., Rangel A., Shimojo S., O'Doherty J. P (2009) Evidence for a Common Representation of Decision Values for Dissimilar Goods in Human Ventromedial Prefrontal Cortex. *Journal of Neuroscience* **29**:12315-12320 <https://doi.org/10.1523/JNEUROSCI.2575-09.2009> | PubMed
- Cisek P., Puskas G. A., El-Murr S (2009) Decisions in Changing Conditions: The Urgency-Gating Model. *Journal of Neuroscience* **29**:11560-11571 <https://doi.org/10.1523/JNEUROSCI.1844-09.2009> | PubMed
- Clithero J. A (2018) Improving out-of-sample predictions using response times and a model of the decision process. *Journal of Economic Behavior & Organization* **148**:344-375 <https://doi.org/10.1016/j.jebo.2018.02.007>
- Desikan R. S., Ségonne F., Fischl B., Quinn B. T., Dickerson B. C., Blacker D., Buckner R. L., Dale A. M., Maguire R. P., Hyman B. T., et al. (2006) An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. *NeuroImage* **31**:968-980 <https://doi.org/10.1016/j.neuroimage.2006.01.021> | PubMed
- Diederich A (1997) Dynamic Stochastic Models for Decision Making under Time Constraints. *Journal of Mathematical Psychology* **41**:260-274 <https://doi.org/10.1006/jmps.1997.1167> | PubMed
- Fiedler S., Glöckner A (2015) Attention and moral behavior. *Current Opinion in Psychology* **6**:139-144 <https://doi.org/10.1016/j.copsyc.2015.08.008>
- Fiedler S., Glöckner A., Nicklisch A., Dickert S (2013) Social Value Orientation and information search in social dilemmas: An eye-tracking analysis. *Organizational Behavior and Human Decision Processes* **120**:272-284 <https://doi.org/10.1016/j.obhdp.2012.07.002>
- Fisher G (2021) Intertemporal Choices Are Causally Influenced by Fluctuations in Visual Attention. *Management Science* **67**:4961-4981 <https://doi.org/10.1287/mnsc.2020.3732>
- Folke T., Jacobsen C., Fleming S. M., De Martino B. (2016) Explicit representation of confidence informs future value-based decisions. *Nature Human Behaviour* **1** <https://doi.org/10.1038/s41562-016-0002>
- Forstmann B. U., Ratcliff R., Wagenmakers E.-J (2016) Sequential Sampling Models in Cognitive Neuroscience: Advantages, Applications, and Extensions. *Annual Review of Psychology* **67**:641-666 <https://doi.org/10.1146/annurev-psych-122414-033645> | PubMed

- Frömer R., Dean Wolf C. K., Shenhav A (2019) Goal congruency dominates reward value in accounting for behavioral and neural correlates of value-based decision-making. *Nature Communications* **10** <https://doi.org/10.1038/s41467-019-12931-x> | PubMed
- Galdo M., Weichart E. R., Sloutsky V. M., Turner B. M (2022) The quest for simplicity in human learning: Identifying the constraints on attention. *Cognitive Psychology* **138**:101508 <https://doi.org/10.1016/j.cogpsych.2022.101508> | PubMed
- Glaholt M. G., Reingold E. M (2009) Stimulus exposure and gaze bias: A further test of the gaze cascade model. *Attention, Perception, & Psychophysics* **71**:445-450 <https://doi.org/10.3758/APP.71.3.445> | PubMed
- Glickman M., Sharoni O., Levy D. J., Niebur E., Stuphorn V., Usher M (2019) The formation of preference in risky choice. *PLOS Computational Biology* **15**:e1007201 <https://doi.org/10.1371/journal.pcbi.1007201> | PubMed
- Gluth S., Rieskamp J., Büchel C (2012) Deciding When to Decide: Time-Variant Sequential Sampling Models Explain the Emergence of Value-Based Decisions in the Human Brain. *Journal of Neuroscience* **32**:10686-10698 <https://doi.org/10.1523/JNEUROSCI.0727-12.2012> | PubMed
- Gluth S., Spektor M. S., Rieskamp J (2018) Value-based attentional capture affects multi-alternative decision making. *eLife* **7**:e39659 <https://doi.org/10.7554/eLife.39659> | PubMed
- Gold J. I., Shadlen M. N (2007) The Neural Basis of Decision Making. *Annual Review of Neuroscience* **30**:535-574 <https://doi.org/10.1146/annurev.neuro.29.051605.113038> | PubMed
- Grinband J., Savitskaya J., Wager T. D., Teichert T., Ferrera V. P., Hirsch J (2011) The dorsal medial frontal cortex is sensitive to time on task, not response conflict or error likelihood. *NeuroImage* **57**:303-311 <https://doi.org/10.1016/j.neuroimage.2010.12.027> | PubMed
- Gwinn R., Leber A. B., Krajbich I (2019) The spillover effects of attentional learning on value-based choice. *Cognition* **182**:294-306 <https://doi.org/10.1016/j.cognition.2018.10.012> | PubMed
- Hanks T. D., Summerfield C (2017) Perceptual Decision Making in Rodents, Monkeys, and Humans. *Neuron* **93**:15-31 <https://doi.org/10.1016/j.neuron.2016.12.003> | PubMed
- Hare T. A., Schultz W., Camerer C. F., O'Doherty J. P., Rangel A (2011) Transformation of stimulus value signals into motor commands during simple choice. *Proceedings of the National Academy of Sciences* **108**:18120-18125 <https://doi.org/10.1073/pnas.1109322108> | PubMed
- Hoffman J. E., Subramaniam B (1995) The role of visual attention in saccadic eye movements. *Perception & Psychophysics* **57**:787-795 <https://doi.org/10.3758/BF03206794> | PubMed
- Holroyd C. B., Ribas-Fernandes J. J. F., Shahnazian D., Silvetti M., Verguts T (2018) Human midcingulate cortex encodes distributed representations of task progress. *Proceedings of the National Academy of Sciences* **115**:6398-6403 <https://doi.org/10.1073/pnas.1803650115> | PubMed
- Hunt L. T., Malalasekera W. N., de Berker A. O., Miranda B., Farmer S. F., Behrens T. E., Kennerley S. W. (2018) Triple dissociation of attention and decision computations across prefrontal cortex. *Nature neuroscience* **21**:1471-1481 <https://doi.org/10.1038/s41593-018-0239-5> | PubMed
- Huys Q. J. M., Cools R., Gölzer M., Friedel E., Heinz A., Dolan R. J., Dayan P (2011) Disentangling the Roles of Approach, Activation and Valence in Instrumental and Pavlovian Responding. *PLOS Computational Biology* **7**:e1002028 <https://doi.org/10.1371/journal.pcbi.1002028> | PubMed
- Jang A. I., Sharma R., Drugowitsch J (2021) Optimal policy for attention-modulated decisions explains human fixation behavior. *eLife* **10**:e63436 <https://doi.org/10.7554/eLife.63436> | PubMed
- Janiszewski C., Kuo A., Tavassoli N. T (2013) The Influence of Selective Attention and Inattention to Products on Subsequent Choice. *Journal of Consumer Research* **39**:1258-1274 <https://doi.org/10.1086/668234>
- Jiang T., Potters J., Funaki Y (2016) Eye-tracking Social Preferences. *Journal of Behavioral Decision Making* **29**:157-168 <https://doi.org/10.1002/bdm.1899>

- Johnson J. G., Busemeyer J. R (2005) A Dynamic, Stochastic, Computational Model of Preference Reversal Phenomena. *Psychological Review* **112**:841 <https://doi.org/10.1037/0033-295X.112.4.841> | PubMed
- Juechems K., Balaguer J., Ruz M., Summerfield C (2017) Ventromedial Prefrontal Cortex Encodes a Latent Estimate of Cumulative Reward. *Neuron* **93**:705-714.e4. <https://doi.org/10.1016/j.neuron.2016.12.038> | PubMed
- Kaanders P., Nili H., O'Reilly J. X., Hunt L (2021) Medial Frontal Cortex Activity Predicts Information Sampling in Economic Choice. *Journal of Neuroscience* **41**:8403-8413 <https://doi.org/10.1523/JNEUROSCI.0392-21.2021> | PubMed
- Kim B., Seligman D., Kable J. W (2012) Preference Reversals in Decision Making Under Risk are Accompanied by Changes in Attention to Different Attributes. *Frontiers in Neuroscience* **6** <https://doi.org/10.3389/fnins.2012.00109> | PubMed
- Kleiner M., Brainard D., Pelli D (2007) What's new in Psychtoolbox-3?. *Perception* **36**:14-14
- Kolling N., Wittmann M. K., Behrens T. E. J., Boorman E. D., Mars R. B., Rushworth M. F. S (2016) Value, search, persistence and model updating in anterior cingulate cortex. *Nature Neuroscience* **19** <https://doi.org/10.1038/nn.4382> | PubMed
- Konovalov A., Krajbich I (2016) Gaze data reveal distinct choice processes underlying model-based and model-free reinforcement learning. *Nature Communications* **7** <https://doi.org/10.1038/ncomms12438> | PubMed
- Konovalov A., Krajbich I (2019) Revealed strength of preference: Inference from response times. *Judgment and Decision Making* **14**:381-394 <https://doi.org/10.1017/s1930297500006082>
- Krajbich I., Armel C., Rangel A (2010) Visual fixations and the computation and comparison of value in simple choice. *Nature Neuroscience* **13**:1292-1298 <https://doi.org/10.1038/nn.2635> | PubMed
- Krajbich I., Rangel A (2011) Multialternative drift-diffusion model predicts the relationship between visual fixations and choice in value-based decisions. *Proceedings of the National Academy of Sciences* **108**:13852-13857 <https://doi.org/10.1073/pnas.1101328108> | PubMed
- Levy D. J., Glimcher P. W (2011) Comparing Apples and Oranges: Using Reward-Specific and Reward-General Subjective Value Representation in the Brain. *Journal of Neuroscience* **31**:14693-14707 <https://doi.org/10.1523/JNEUROSCI.2218-11.2011> | PubMed
- Li Z.-W., Ma W. J (2021) An uncertainty-based model of the effects of fixation on choice. *PLOS Computational Biology* **17**:e1009190 <https://doi.org/10.1371/journal.pcbi.1009190> | PubMed
- Lim S.-L., O'Doherty J. P., Rangel A (2011a) The decision value computations in the vmPFC and striatum use a relative value code that is guided by visual attention. *Journal of Neuroscience* **31**:13214-13223 <https://doi.org/10.1523/JNEUROSCI.1246-11.2011> | PubMed
- Lim S.-L., O'Doherty J. P., Rangel A (2011b) The Decision Value Computations in the vmPFC and Striatum Use a Relative Value Code That is Guided by Visual Attention. *Journal of Neuroscience* **31**:13214-13223 <https://doi.org/10.1523/JNEUROSCI.1246-11.2011> | PubMed
- Lopez-Persem A., Domenech P., Pessiglione M (2016) How prior preferences determine decision-making frames and biases in the human brain. *eLife* **5**:e20317 <https://doi.org/10.7554/eLife.20317> | PubMed
- MathWorks (2016) MATLAB. version: R2016b
- McGinty V. B (2019) Overt attention toward appetitive cues enhances their subjective value, independent of orbitofrontal cortex activity. *eNeuro* **6** <https://doi.org/10.1523/ENEURO.0230-19.2019> | PubMed
- McGinty V. B., Rangel A., Newsome W. T (2016) Orbitofrontal cortex value signals depend on fixation location during free viewing. *Neuron* **90**:1299-1311 <https://doi.org/10.1016/j.neuron.2016.04.045> | PubMed
- Mullett T. L., Stewart N (2016) Implications of visual attention phenomena for models of preferential choice. *Decision* **3**:231 <https://doi.org/10.1037/dec0000049> | PubMed

- Mumford J. A., Bissett P. G., Jones H. M., Shim S., Rios J. A. H., Poldrack R. A (2024) The response time paradox in functional magnetic resonance imaging analyses. *Nature Human Behaviour* **8**:349-360 <https://doi.org/10.1038/s41562-023-01760-0> | PubMed
- Nichols T. E., Holmes A. P (2002) Nonparametric permutation tests for functional neuroimaging: A primer with examples. *Human Brain Mapping* **15**:1-25 <https://doi.org/10.1002/hbm.1058> | PubMed
- O’Connell R. G., Shadlen M. N., Wong-Lin K., Kelly S. P (2018) Bridging Neural and Computational Viewpoints on Perceptual Decision-Making. *Trends in Neurosciences* **41**:838-852 <https://doi.org/10.1016/j.tins.2018.06.005> | PubMed
- Orquin J. L., Mueller Loose S (2013) Attention and choice: A review on eye movements in decision making. *Acta Psychologica* **144**:190-206 <https://doi.org/10.1016/j.actpsy.2013.06.003> | PubMed
- Pärnamets P., Johansson P., Hall L., Balkenius C., Spivey M. J., Richardson D. C (2015) Biasing moral decisions by exploiting the dynamics of eye gaze. *Proceedings of the National Academy of Sciences* **112**:4170-4175 <https://doi.org/10.1073/pnas.1415250112> | PubMed
- Pelli D. G (1997) The VideoToolbox software for visual psychophysics: Transforming numbers into movies. *Spatial Vision* **10**:437-442 <https://doi.org/10.1163/156856897X00366> | PubMed
- Pisauro M. A., Fouragnan E., Retzler C., Philiastides M. G (2017) Neural correlates of evidence accumulation during value-based decisions revealed via simultaneous EEG-fMRI. *Nature Communications* **8** <https://doi.org/10.1038/ncomms15808> | PubMed
- Pleskac T. J., Yu S., Grunevski S., Liu T (2022) Attention biases preferential choice by enhancing an option’s value. *Journal of Experimental Psychology: General* <https://doi.org/10.1037/xge0001307> | PubMed
- Polanía R., Krajbich I., Grueschow M., Ruff C. C (2014) Neural oscillations and synchronization differentially support evidence accumulation in perceptual and value-based decision making. *Neuron* **82**:709-720 <https://doi.org/10.1016/j.neuron.2014.03.014> | PubMed
- Polonio L., Di Guida S., Coricelli G. (2015) Strategic sophistication and attention in games: An eye-tracking study. *Games and Economic Behavior* **94**:80-96 <https://doi.org/10.1016/j.geb.2015.09.003>
- Pooley J. P., Lee M. D., Shanks W. R (2011) Understanding memory impairment with memory models and hierarchical Bayesian analysis. *Journal of Mathematical Psychology* **55**:47-56 <https://doi.org/10.1016/j.jmp.2010.08.003>
- Ratcliff R (1978) A theory of memory retrieval. *Psychological Review* **85**:59 <https://doi.org/10.1037/0033-295x.85.2.59>
- Ratcliff R., Smith P. L (2004) A comparison of sequential sampling models for two-choice reaction time. *Psychological Review* **111**:333 <https://doi.org/10.1037/0033-295X.111.2.333> | PubMed
- Ratcliff R., Smith P. L., Brown S. D., McKoon G (2016) Diffusion Decision Model: Current issues and history. *Trends in Cognitive Sciences* **20**:260-281 <https://doi.org/10.1016/j.tics.2016.01.007> | PubMed
- Rich E. L., Wallis J. D (2016) Decoding subjective decisions from orbitofrontal cortex. *Nature Neuroscience* **19** <https://doi.org/10.1038/nn.4320> | PubMed
- Rodriguez C. A., Turner B. M., Zandt T. V., McClure S. M (2015) The neural basis of value accumulation in intertemporal choice. *European Journal of Neuroscience* **42**:2179-2189 <https://doi.org/10.1111/ejn.12997> | PubMed
- Roe R. M., Busemeyer J. R., Townsend J. T (2001) Multialternative decision field theory: A dynamic connectionst model of decision making. *Psychological Review* **108**:370-392 <https://doi.org/10.1037/0033-295X.108.2.370> | PubMed
- Rouault M., Drugowitsch J., Koechlin E (2019) Prefrontal mechanisms combining rewards and beliefs in human decision-making. *Nature Communications* **10** <https://doi.org/10.1038/s41467-018-08121-w> | PubMed
- Russo J. E., Leclerc F (1994) An Eye-Fixation Analysis of Choice Processes for Consumer Nondurables. *Journal of Consumer Research* **21**:274-290 <https://doi.org/10.1086/209397>

- Sepulveda P., Usher M., Davies N., Benson A. A., Ortoleva P., De Martino B. (2020) Visual attention modulates the integration of goal-relevant evidence and not value. *eLife* **9**:e60705 <https://doi.org/10.7554/eLife.60705> | PubMed
- Shadlen M. N., Shohamy D (2016) Decision making and sequential sampling from memory. *Neuron* **90**:927-939 <https://doi.org/10.1016/j.neuron.2016.04.036> | PubMed
- Sheng F., Ramakrishnan A., Seok D., Zhao W. J., Thelaus S., Cen P., Platt M. L (2020) Decomposing loss aversion from gaze allocation and pupil dilation. *Proceedings of the National Academy of Sciences* **117**:11356-11363 <https://doi.org/10.1073/pnas.1919670117> | PubMed
- Shenhav A., Straccia M. A., Botvinick M. M., Cohen J. D (2016) Dorsal anterior cingulate and ventromedial prefrontal cortex have inverse roles in both foraging and economic choice. *Cognitive, Affective, & Behavioral Neuroscience* **16**:1127-1139 <https://doi.org/10.3758/s13415-016-0458-8> | PubMed
- Shenhav A., Straccia M. A., Cohen J. D., Botvinick M. M (2014) Anterior cingulate engagement in a foraging context reflects choice difficulty, not foraging value. *Nature Neuroscience* **17** <https://doi.org/10.1038/nn.3771> | PubMed
- Shi S. W., Wedel M., Pieters F. G. M. (2012) Information Acquisition During Online Decision Making: A Model-Based Exploration Using Eye-Tracking Data. *Management Science* **59**:1009-1026 <https://doi.org/10.1287/mnsc.1120.1625>
- Shimojo S., Simion C., Shimojo E., Scheier C (2003) Gaze bias both reflects and influences preference. *Nature Neuroscience* **6** <https://doi.org/10.1038/nn1150> | PubMed
- Smith S. M., Krajbich I (2018) Attention and choice across domains. *Journal of Experimental Psychology: General* **147**:1810-1826 <https://doi.org/10.1037/xge0000482> | PubMed
- Smith S. M., Krajbich I (2019) Gaze Amplifies Value in Decision Making. *Psychological Science* **30**:116-128 <https://doi.org/10.1177/0956797618810521> | PubMed
- Smith S. M., Krajbich I., Webb R (2019) Estimating the dynamic role of attention via random utility. *Journal of the Economic Science Association* **5**:97-111 <https://doi.org/10.1007/s40881-019-00062-4>
- Stewart N., Hermens F., Matthews W. J (2016) Eye movements in risky choice. *Journal of Behavioral Decision Making* **29**:116-136 <https://doi.org/10.1002/bdm.1854> | PubMed
- Tavares G., Perona P., Rangel A (2017) The attentional drift diffusion model of simple perceptual decision-making. *Frontiers in Neuroscience* **11** <https://doi.org/10.3389/fnins.2017.00468> | PubMed
- Teoh Y. Y., Yao Z., Cunningham W. A., Hutcherson C. A (2020) Attentional priorities drive effects of time pressure on altruistic choice. *Nature Communications* **11**:3534 <https://doi.org/10.1038/s41467-020-17326-x> | PubMed
- Thomas A. W., Molter F., Krajbich I., Heekeren H. R., Mohr P. N. C (2019) Gaze bias differences capture individual choice behaviour. *Nature Human Behaviour* **3** <https://doi.org/10.1038/s41562-019-0584-8> | PubMed
- Towal R. B., Mormann M., Koch C (2013) Simultaneous modeling of visual saliency and value computation improves predictions of economic choice. *Proceedings of the National Academy of Sciences of the United States of America* **110**:E3858-E3867 <https://doi.org/10.1073/pnas.1304429110> | PubMed
- Tsetos K., Chater N., Usher M (2012) Saliency driven value integration explains decision biases and preference reversal. *Proceedings of the National Academy of Sciences* **109**:9659-9664 <https://doi.org/10.1073/pnas.1119569109> | PubMed
- Usher M., McClelland J. L (2001) The time course of perceptual choice: The leaky, competing accumulator model. *Psychological Review* **108**:550 <https://doi.org/10.1037/0033-295x.108.3.550> | PubMed
- Vaidya A. R., Fellows L. K (2015) Testing necessary regional frontal contributions to value assessment and fixation-based updating. *Nature Communications* **6** <https://doi.org/10.1038/ncomms10120> | PubMed

- Vanunu Y., Hotaling J. M., Le Pelley M. E., Newell B. R. (2021) How top-down and bottom-up attention modulate risky choice. *Proceedings of the National Academy of Sciences* **118**:e2025646118 <https://doi.org/10.1073/pnas.2025646118> | PubMed
- Vassena E., Deraeve J., Alexander W. H (2020) Surprise, value and control in anterior cingulate cortex during speeded decision-making. *Nature Human Behaviour* **4** <https://doi.org/10.1038/s41562-019-0801-5> | PubMed
- Venkatraman V., Payne J. W., Huettel S. A (2014) An overall probability of winning heuristic for complex risky decisions: Choice and eye fixation evidence. *Organizational Behavior and Human Decision Processes* **125**:73-87 <https://doi.org/10.1016/j.obhdp.2014.06.003>
- Wang J. T., Spezio M., Camerer C. F (2010) Pinocchio's Pupil: Using Eyetracking and Pupil Dilation to Understand Truth Telling and Deception in Sender-Receiver Games. *American Economic Review* **100**:984-1007 <https://doi.org/10.1257/aer.100.3.984>
- Wang X.-J (2002) Probabilistic Decision Making by Slow Reverberation in Cortical Circuits. *Neuron* **36**:955-968 [https://doi.org/10.1016/S0896-6273\(02\)01092-9](https://doi.org/10.1016/S0896-6273(02)01092-9)
- Westbrook A., van den Bosch R., Määttä J. I., Hofmans L., Papadopetraki D., Cools R., Frank M. J. (2020) Dopamine promotes cognitive effort by biasing the benefits versus costs of cognitive work. *Science* **367**:1362-1366 <https://doi.org/10.1126/science.aaz5891> | PubMed
- Wittmann M. K., Fouragnan E., Folloni D., Klein-Flügge M. C., Chau B. K. H., Khamassi M., Rushworth M. F. S (2020) Global reward state affects learning and activity in raphe nucleus and anterior insula in monkeys. *Nature Communications* **11** <https://doi.org/10.1038/s41467-020-17343-w> | PubMed
- Wittmann M. K., Kolling N., Akaishi R., Chau B. K. H., Brown J. W., Nelissen N., Rushworth M. F. S (2016) Predictive decision making driven by multiple time-linked reward representations in the anterior cingulate cortex. *Nature Communications* **7** <https://doi.org/10.1038/ncomms12327> | PubMed
- Yang T., Shadlen M. N (2007) Probabilistic reasoning by neurons. *Nature* **447** <https://doi.org/10.1038/nature05852> | PubMed
- Yang X., Krajbich I (2023) A dynamic computational model of gaze and choice in multi-attribute decisions. *Psychological Review* **130**:52-70 <https://doi.org/10.1037/rev0000350> | PubMed
- Zilker V., Pachur T (2023) Attribute attention and option attention in risky choice. *Cognition* **236**:105441 <https://doi.org/10.1016/j.cognition.2023.105441> | PubMed
- Zilker V., Pachur T (2023) Nonlinear probability weighting can reflect attentional biases in sequential sampling. *Psychological Review* **129**:949 <https://doi.org/10.1037/rev0000304> | PubMed
- Shevlin, Gwinn, Makwana, Krajbich (2026) Behavioral data from: Overt visual attention modulates decision-related signals in ventral and dorsal medial prefrontal cortex. Open Science Framework. ID eyxvb <https://osf.io/eyxvb/>
- Shevlin, Gwinn, Makwana, Krajbich (2026) fMRI data from: Overt visual attention modulates decision-related signals in ventral and dorsal medial prefrontal cortex. OpenNeuro. <https://doi.org/10.18112/openneuro.ds007305.v1.0.0>

Peer reviews

Reviewer #1 (Public review):

Summary:

This study builds upon a major theoretical account of value-based choice, the 'attentional drift diffusion model' (aDDM), and examines whether and how this might be implemented in the human brain using functional magnetic resonance imaging (fMRI). The aDDM states that the process of internal evidence accumulation across time should be weighted by the decision maker's gaze, with more weight being assigned to the currently fixated item. The present study aims to test whether there are (a) regions of the brain where signals related to the

currently presented value are affected by the participant's gaze; (b) regions of the brain where previously accumulated information is weighted by gaze.

To examine this, the authors developed a novel paradigm that allowed them to dissociate currently and previously presented evidence, at a timescale amenable to measuring neural responses with fMRI. They asked participants to choose between bundles or 'lotteries' of food times, which they revealed sequentially and slowly to the participant across time. This allowed modelling of the haemodynamic response to each new observation in the lottery, separately for previously accumulated and currently presented evidence.

Using this approach, they find that regions of the brain supporting valuation (vmPFC and ventral striatum) have responses reflecting gaze-weighted valuation of the currently presented item, whereas regions previously associated with evidence accumulation (preSMA and IPS) have responses reflected gaze-weighted modulation of previously accumulated evidence.

A major strength of the current paper is the design of the task, nicely allowing the researchers to examine evidence accumulation across time despite using a technique with poor temporal resolution. The dissociation between currently presented and previously accumulated evidence in different brain regions in GLM1 (before gaze-weighting), as presented in Figure 5, is already compelling. The result that regions such as preSMA response positively to $|AV|$ (absolute difference in accumulated value) is particularly interesting, as it would seem that the 'decision conflict' account of this region's activity might predict the exact opposite result. Additionally, the behaviour has been well modelled at the end of the paper when examining temporal weighting functions across the multiple samples.

In response to reviewer comments, the authors have explicitly tested for the effects of gaze-weighting over and above any main effect of value, and convincingly shown that these effects are both present in the main regions of interest - namely $|SV|$ and gaze-weighted $|SV|$ in the vmPFC, alongside $|AV|$ and $|AV_gaze|$ in the pre-SMA. This provides clear evidence in support of the notion of gaze-weighting of value signals in these regions.

<https://doi.org/10.7554/eLife.103846.2.sa2>

Reviewer #2 (Public review):

Summary:

In this paper the authors seek to disentangle brain areas that encode the subjective value of individual stimuli/items (input regions) from those that accumulate those values into decision variables (integrators) for value-based choice. The authors used a novel task in which stimulus presentation was slowed down to ensure that such a dissociation was possible using fMRI despite its relatively low temporal resolution. In addition, the authors leveraged the fact that gaze increases item value, providing a means of distinguishing brain regions that encode decision variables from those that encode other quantities such as conflict or time-on-task. The authors adopt a region-of-interest approach based on an extensive previous literature and found that the ventral striatum and vmPFC correlated with the item values and not their accumulation whereas the pre-SMA, IPS and dlPFC correlated more strongly with their accumulation. Further analysis revealed that the pre-SMA was the only one of the three integrator regions to also exhibit gaze modulation.

The study uses a highly innovative design and addresses an important and timely topic. The manuscript is well-written and engaging, while the data analysis appears highly rigorous.

Weaknesses:

With 23 subjects the study has relatively low statistical power for fMRI although the within-subjects design and relatively high trial count reduces these concerns.

<https://doi.org/10.7554/eLife.103846.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

This study builds upon a major theoretical account of value-based choice, the 'attentional drift diffusion model' (aDDM), and examines whether and how this might be implemented in the human brain using functional magnetic resonance imaging (fMRI). The aDDM states that the process of internal evidence accumulation across time should be weighted by the decision maker's gaze, with more weight being assigned to the currently fixated item. The present study aims to test whether there are (a) regions of the brain where signals related to the currently presented value are affected by the participant's gaze; (b) regions of the brain where previously accumulated information is weighted by gaze.

To examine this, the authors developed a novel paradigm that allowed them to dissociate currently and previously presented evidence, at a timescale amenable to measuring neural responses with fMRI. They asked participants to choose between bundles or 'lotteries' of food times, which they revealed sequentially and slowly to the participant across time. This allowed modelling of the haemodynamic response to each new observation in the lottery, separately for previously accumulated and currently presented evidence.

Using this approach, they find that regions of the brain supporting valuation (vmPFC and ventral striatum) have responses reflecting gaze-weighted valuation of the currently presented item, whereas regions previously associated with evidence accumulation (preSMA and IPS) have responses reflecting gaze-weighted modulation of previously accumulated evidence.

Strengths:

A major strength of the current paper is the design of the task, nicely allowing the researchers to examine evidence accumulation across time despite using a technique with poor temporal resolution. The dissociation between currently presented and previously accumulated evidence in different brain regions in GLM1 (before gaze-weighting), as presented in Figure 5, is already compelling. The result that regions such as preSMA respond positively to $|AV|$ (absolute difference in accumulated value) is particularly interesting, as it would seem that the 'decision conflict' account of this region's activity might predict the exact opposite result. Additionally, the behaviour has been well modelled at the end of the paper when examining temporal weighting functions across the multiple samples.

Weaknesses:

The results relating to gaze-weighting in the fMRI signal could do with some further explication to become more complete. A major concern with GLM2, which looks at the same effects as GLM1 but now with gaze-weighting, is that these gaze-weighted

regressors may be (at least partially) correlated with their non-gaze-weighted counterparts (e.g., SVgaze will correlate with SV). But the non-gaze-weighted regressors have been excluded from this model. In other words, the authors are not testing for effects of gaze-weighting of value signals *over and above* the base effects of value in this model. In my mind, this means that the GLM2 results could simply be a replication of the findings from GLM1 at present. GLM3 is potentially a stronger test, as it includes the value signals and the interaction with gaze in the same model. But here, while the link to the currently attended item is quite clear (and a replication of Lim et al, 2011), the link to previously accumulated evidence is a bit contorted, depending upon the interpretation of a behavioural regression to interpret the fMRI evidence. The results from GLM3 are also, by the authors' own admission, marginal in places.

We have addressed this comment with new GLMs. The new GLM1 includes both non-gazeweighted and gaze-weighted regressors and finds that the vmPFC and striatum reflect gazeweighted sampled value, while the preSMA reflects gaze-weighted accumulated value. We have now dropped the old GLM3 and added two other GLMs, one that explicitly interacts accumulated value with accumulated dwell, and the other that considers only partial gaze discounting. These analyses all support the preSMA as encoding gaze-weighted accumulated value.

Reviewer #2 (Public review):

Summary:

In this paper, the authors seek to disentangle brain areas that encode the subjective value of individual stimuli/items (input regions) from those that accumulate those values into decision variables (integrators) for value-based choice. The authors used a novel task in which stimulus presentation was slowed down to ensure that such a dissociation was possible using fMRI despite its relatively low temporal resolution. In addition, the authors leveraged the fact that gaze increases item value, providing a means of distinguishing brain regions that encode decision variables from those that encode other quantities such as conflict or time-on-task. The authors adopt a region-of-interest approach based on an extensive previous literature and found that the ventral striatum and vmPFC correlated with the item values and not their accumulation, whereas the pre-SMA, IPS, and dlPFC correlated more strongly with their accumulation. Further analysis revealed that the preSMA was the only one of the three integrator regions to also exhibit gaze modulation.

Strengths:

The study uses a highly innovative design and addresses an important and timely topic. The manuscript is well-written and engaging, while the data analysis appears highly rigorous.

Weaknesses:

With 23 subjects, the study has relatively low statistical power for fMRI.

We believe several features of our study design and analytic approach mitigate concerns regarding statistical power.

First, our paradigm leveraged a within-subjects design with high total sample counts. Each participant completed approximately 60 choice trials across three 15-minute runs, with an average of 6.37 samples per trial. This yielded roughly 380 observations per participant, providing substantial statistical power at the individual level before aggregating across subjects. This within-subject power is particularly important for detecting parametric effects, as our regressors of interest ($|\Delta SV|$ and $|\Delta AV|$) varied continuously across and within trials.

Second, rather than conducting an exploratory whole-brain analysis that would require larger sample sizes to correct for multiple comparisons, we employed a targeted ROI approach based on well-established regions from prior literature (e.g., Bartra et al., 2013; Hare et al., 2011). This ROI-driven approach substantially increases statistical power by reducing the search space and leverages theoretical predictions about where effects should occur. Our novel contribution that gaze modulation of accumulated evidence signals was reflected in preSMA activity builds naturally on established findings. However, we acknowledge that a larger sample size would provide greater confidence in the null effects and would enable more detailed individual differences analyses.

We have added a brief acknowledgement of the sample size limitation to the Discussion section of the main text:

“While our sample size of 20 subjects is modest by current neuroimaging standards, the withinsubject statistical power from our extended decision paradigm (~380 observations per subject), combined with hypothesis-driven ROI analyses and multiple comparisons correction, provides confidence in our core findings. Nevertheless, replication with larger samples would be valuable, particularly for more fully characterizing null effects and marginal findings.”

Recommendations for the authors:

Editor Comments:

Reviewer 1 in particular makes a number of suggestions for additional analyses that would help to strengthen the evidence supporting your conclusions.

We thank the editor and the reviewers for the helpful suggestions for improving our manuscript. We discuss our efforts to address each point below.

Reviewer #1 (Recommendations for the authors):

(1) To address my concerns about GLM2, the first thing to do might be to simply show the correlation between the regressors used across the three different models (e.g., as a figure in the methods). Although the authors have done a good job to ensure that AV and SV are decorrelated when including them both in the same model, they haven't shown us whether the regressors used in, for example, GLM2 are correlated/similar to the regressors used in GLM1. This is important information for interpretation.

Thank you for raising concerns about the overlap between different models. We agree that additional information regarding the correlation among sample-level regressors would aid readers in understanding the differences among the analyses. We now include this information in Figure 7 in the Methods section, as requested. While $|SV|$ was uncorrelated with gaze-weighted $|SV|$ ($|SV_{Gaze}|$; Pearson's $r = 0.002$, $p = 0.848$), lagged $|AV|$ was significantly correlated with lagged, gaze-weighted $|AV|$ (lagged $|AV_{Gaze}|$; $r = 0.365$, $p < 2.2 \times 10^{-16}$).

*(2) The acid test for gaze-modulation of value signals would be to show that the gazemodulated signals explain the fMRI results over and above the non-gaze-modulated signals. This could simply mean including SVgaze and SV (and equivalent terms for AV) within the same GLM. Following from point (1), the authors may point out that these terms are highly correlated - yes, but the GLM will then test for the effects of SVgaze *over and above* the effects of SV. (In fact, although I'd normally caution against orthogonalisation - it would here be totally legitimate to orthogonalise SVgaze w.r.t. SV).*

We appreciate the reviewer's suggestions for more robust tests of the presence of gaze-weighted signals. For reasons highlighted in our response above, we were initially hesitant to

include both types of regressors in the same model due to their significant correlation. However, we now report the results of this analysis in the main text as the new GLM 1. This model incorporates both gaze-weighted and non-gaze-weighted terms. For each contrast we used the same procedures as reported in the main text (family-wise error corrected at $p < 0.05$ and clusterforming thresholds at $p < 0.005$).

In the vmPFC, we found significant effects of both $|\Delta SV|$ (peak voxel: $x = -14, y = 44, z = -12$; $t = 3.90, p = 0.0190$) and $|\Delta SV_{\text{Gaze}}|$ (peak voxel: $x = 4, y = 38, z = -4$; $t = 5.21, p = 0.004$), but no effects of $|\Delta AV|$ or $|\Delta AV_{\text{Gaze}}|$. The striatum also showed a significant correlation with $|\Delta SV_{\text{Gaze}}|$ (peak voxel: $x = 22, y = 20, z = -10$; $t = 5.10, p = 0.014$), but no other regressors.

In the pre-SMA, we found a significantly positive relationship with both $|\Delta AV|$ (peak voxel: $x = 4, y = 14, z = 50$; $t = 4.75, p < 0.001$) and $|\Delta AV_{\text{Gaze}}|$ (peak voxel: $x = 4, y = 18, z = 50$; $t = 2.98, p = 0.032$). In contrast, the dlPFC ($x = 40, y = 34, z = 26$; $t = 6.83, p < 0.001$) and IPS ($x = 42, y = -50, z = 42$; $t = 5.16, p = 0.010$) were only correlated with $|\Delta AV|$. No other significant contrasts emerged.

These results provide direct support for the presence of gaze-modulated value signals in the brain, which we now describe in the main text Results section.

(3) With regards to GLM3, it would help to provide a bit more detail on what the time series looks like for the gaze regressor in this model - is it the entire timeseries of gaze (which presumably shifts back/forth between options multiple times within each trial) which is being convolved with the HRF? This seems different from how gaze is being calculated in GLM2, where it is amalgamated into an 'average gaze difference' within a sample between left/right options, if I understand the text correctly?

We apologize for the lack of details regarding how we operationalized the gaze regressors in our analyses. You are correct that the gaze regressor was calculated differently in GLM2 and GLM3.

However, in response to the reviewer's points above (Major Point 2) and below (Major Point 4, Minor Point 1), we have decided to drop the old GLM3 from the paper while incorporating a revised GLM1 (combining old GLM1 and GLM2) and two new GLMs (see responses to Major Point 4 and Minor Point 1) to provide clearer evidence for gaze modulation of accumulated value in the brain.

*(4) Also, is there not a reason why it isn't more appropriate to interact AV with *previously deployed gaze difference* (accumulated across previous samples) in this model, rather than the current gaze location? The latter seems to rely upon the indirect linkage via the behavioural modelling result, which seems to weaken the claim.*

We thank the reviewer for this suggestion. We agree that our original GLM3 approach was limited because it interacted AV with current binary gaze location, which relies on the indirect behavioral relationship we established (i.e., that current gaze is negatively correlated with accumulated past gaze).

The original GLM2 (which is now incorporated into the new GLM1) implemented something similar to what the reviewer is suggesting as it used gaze-weighted values accumulated across all previous samples. Specifically, in GLM2, the gaze-weighted accumulated value (AV_{gaze}) was calculated as the sum of all previous sampled values, each weighted by the proportion of gaze allocated to each option during that sampling period.

However, to more directly test whether accumulated evidence signals are modulated by accumulated gaze allocation we have now run an additional analysis (GLM2). In this analysis we have revised the old GLM3 to include additional regressors: ΔSV , lagged ΔAV , current gaze

location, accumulated dwell advantage, $\Delta SV \times$ current gaze location, and lagged $\Delta AV \times$ accumulated dwell advantage.

The two new regressors were defined as follows:

Accumulated dwell advantage: For each sample t , accumulated dwell advantage represents the cumulative difference in gaze allocation up to sample $t-1$, calculated as (total dwell left – total dwell right) / (total dwell left + total dwell right). This is a continuous measure from -1 (all previous gaze to right) to +1 (all previous gaze to left).

$\Delta AV \times$ accumulated dwell advantage: The interaction between accumulated values and accumulated dwell advantage, which directly tests whether brain regions encoding accumulated value are modulated by the history of gaze allocation.

This approach is conceptually similar to old GLM2's gaze-weighting method, but allows us to examine the interaction effect more explicitly as a separate regressor rather than having it embedded within the value calculation.

Here, we found that the pre-SMA showed a positive correlation with the $\Delta AV \times$ accumulated dwell advantage term (peak voxel: $x = 8, y = 10, z = 58; t = 3.10, p = 0.0258$). Surprisingly, the striatum also showed a correlation with this term (peak: $x = -16, y = 10, z = -6; t = 4.07, p = 0.0176$). No other ROIs showed significant relationships.

This analysis provides additional evidence that pre-SMA encodes accumulated value signals that are modulated by accumulated gaze allocation, without relying on indirect relationships between current and past gaze. We now report these results in the main text as GLM2 as follows:

“To more directly test whether accumulated evidence signals were modulated by accumulated gaze allocation throughout a trial, we conducted additional, exploratory analyses. Specifically, we ran a GLM that incorporated the following two terms: accumulated dwell advantage and $\Delta AV \times$ accumulated dwell advantage, in addition to ΔSV , the current gaze location, and $\Delta SV \times$ current gaze location.

We calculated accumulated dwell advantage as follows: For each sample t , accumulated dwell advantage is the cumulative difference in gaze allocation up to sample $t-1$, calculated as (total dwell left – total dwell right) / (total dwell left + total dwell right). This is a continuous measure from -1 (all previous gaze to right) to +1 (all previous gaze to left).

We also included the interaction between accumulated dwell advantage and ΔAV (i.e., signed accumulated evidence). This interaction term is positive when gaze is primarily to the left and left has more value or when gaze is primarily to the right and right has more value. This interaction term directly tests whether brain regions encoding accumulated evidence are modulated by the history of gaze allocation. This approach allows us to examine the interaction effect more explicitly as a separate regressor rather than having it embedded within the value calculation itself.

This GLM revealed a positive correlation between pre-SMA activity and the $\Delta AV \times$ accumulated dwell advantage term (peak voxel: $x = 8, y = 10, z = 58; t = 3.01, p = 0.026$). Surprisingly, the striatum also showed a correlation with this term (peak voxel: $x = -16, y = 10, z = -6; t = 4.07, p = 0.018$). Additionally, activity in the dlPFC was positively correlated with ΔSV (peak voxel: $x = -36, y = 34, z = 22; t = 3.96, p = 0.016$). No other ROIs showed significant relations.

This analysis provides additional evidence that the pre-SMA encodes accumulated value signals that are modulated by the history of gaze allocation.”

Minor

(1) "In Trial A, the subject looks left 30% of the time and right 70% of the time. In Trial B, the subject looks left 70% of the time and right 30% of the time. In Trial A, the net input value ("drift rate") would be $|0.3 \cdot 7 - 0.7 \cdot 3| = 0$. In Trial B, the drift rate would be $|0.7 \cdot 7 - 0.3 \cdot 3| = 4$." I may be missing something, but isn't this consistent with an aDDM with $\theta=0$, rather than $\theta=0.3-0.5$ as is typically found?

The reviewer raises an important point about our assumptions regarding attentional discounting. We agree that our approach could be problematic as it may assume stronger discounting than has been observed in the literature.

To address this concern, we calculated drift on a sample-by-sample basis before aggregating to the trial level. Following Smith, Krajbich, and Webb (2019), for each individual sample within a trial, we computed:

$$\beta = (G_{\text{Left}} \times V_{\text{Left}}) - (G_{\text{Right}} \times V_{\text{Right}})$$

$$\gamma = (G_{\text{Right}} \times V_{\text{Left}}) - (G_{\text{Left}} \times V_{\text{Right}}),$$

where G_{Left} and G_{Right} represent the proportion of time spent fixating left versus right within that specific sample, and V_{Left} and V_{Right} are the instantaneous values of the left and right options. We then averaged these sample-level β and γ values across all samples within each trial to obtain trial-level regressors. This approach preserves the fine-grained temporal dynamics of gazedependent value accumulation that would be lost by calculating gaze proportions only at the trial level.

Using this sample-level method in a mixed-effects logistic regression predicting choice (left vs. right), we estimated subject-specific values of $\theta = \gamma/\beta$. Across our sample ($N=20$), we found mean $\theta = 0.77$ ($SD = 0.21$, range = 0.55–1.25). These estimates are somewhat higher than the typical aDDM findings of attentional bias ($\theta = 0.3-0.5$). This may reflect the drawn-out nature of this task relative to prior aDDM tasks.

Next, we ran a new GLM that incorporated these θ estimates in the sampled value estimates. For this GLM3, we computed θ -weighted sampled-value ($|\Delta TWSV|$) as:

$$TWSV = (G_{\text{Left}} \times (V_{\text{Left}} - \theta V_{\text{Right}})) - (G_{\text{Right}} \times (V_{\text{Right}} - \theta V_{\text{Left}})).$$

Similar to GLM1, we computed an accumulated value signal based on the lagged sum of previous samples' $|\Delta TWSV|$ (i.e., $|\Delta TWAV|$).

We found significant positive effects of $|\Delta TWSV|$ in the vmPFC (peak voxel: $x = -14$, $y = 44$, $z = -12$; $t = 3.57$, $p = 0.0270$) and IPS (peak voxel: $x = 30$, $y = -28$, $z = 40$; $t = 4.58$, $p = 0.0198$), but in no other ROI.

In contrast, we found significant positive relationships between $|\Delta TWAV|$ and activity in the preSMA (peak voxel: $x = 0$, $y = 22$, $z = 52$; $t = 4.68$, $p = 0.0014$), dlPFC (peak voxel: $x = 40$, $y = 32$, $z = 26$; $t = 4.32$, $p = 0.0040$), and IPS (peak voxel: $x = 44$, $y = -48$, $z = 42$; $t = 6.26$, $p < 0.0000$). Notably, we also observed a significant relationship between $|\Delta TWAV|$ and activity in the vmPFC ($x = 8$, $y = 38$, $z = 18$; $t = 3.89$, $p = 0.0410$). No other significant contrasts emerged.

We now report this additional analysis as GLM3 in the main text, as follows:

"In our first set of analyses, we implicitly assumed complete discounting of non-fixated information, in contrast with previous studies that have generally found only partial discounting (Krajbich et al., 2010; Sepulveda et al., 2020; Smith & Krajbich, 2019; Westbrook et al., 2020). To verify that our results are robust to inter-subject variability in attentional

discounting, we estimated subject-level attentional discounting parameters and then re-estimated our original GLM with new, recalculated gaze-weighted value regressors.

Following Smith, Krajbich, and Webb (2019), for each individual sample within a trial, we computed:

$\beta = (G_{\text{Left}} \times V_{\text{Left}}) - (G_{\text{Right}} \times V_{\text{Right}})$ $\gamma = (G_{\text{Right}} \times V_{\text{Left}}) - (G_{\text{Left}} \times V_{\text{Right}})$, where G_{Left} and G_{Right} represent the proportion of time spent gazing left versus right within that specific sample, and V_{Left} and V_{Right} are the instantaneous values of the left and right options. We then averaged these sample-level β and γ values across all samples within each trial to obtain trial-level regressors. We then ran a mixed-effects logistic regression predicting choice (left vs. right) as a function of β and γ and then calculated subject-specific values of $\theta = \gamma/\beta$. Across our sample ($N=20$), we found mean $\theta = 0.77$ (SD = 0.21, range = 0.55–1.25).

Next, for the GLM, we computed θ -weighted sampled-value ($|\Delta SV_{\theta}|$) as:

$$\Delta SV_{\theta} = (G_{\text{Left}} \times (V_{\text{Left}} - \theta V_{\text{Right}})) - (G_{\text{Right}} \times (V_{\text{Right}} - \theta V_{\text{Left}}))$$

Similar to the original GLM, we computed an accumulated value signal, $|\Delta AV_{\theta}|$, based on the lagged sum of previous samples' $|\Delta SV_{\theta}|$.

We found significant positive effects of $|\Delta SV_{\theta}|$ in the vmPFC (peak voxel: $x = -14$, $y = 44$, $z = 12$; $t = 3.57$ $p = 0.027$) and IPS (peak voxel: $x = 30$, $y = -28$, $z = 40$; $t = 4.58$ $p = 0.020$), but in no other ROI.

In contrast, we found significant positive relationships between $|\Delta AV_{\theta}|$ and activity in the preSMA (peak voxel: $x = 0$, $y = 22$, $z = 52$; $t = 4.68$, $p = 0.001$), dlPFC (peak voxel: $x = 40$, $y = 32$, $z = 26$; $t = 4.32$, $p = 0.004$), and IPS (peak voxel: $x = 44$, $y = -48$, $z = 42$; $t = 6.26$, $p < 0.0001$). Notably, we also observed a significant relationship between $|\Delta AV_{\theta}|$ and activity in the vmPFC ($x = 8$, $y = 38$, $z = 18$; $t = 3.89$, $p = 0.041$). No other significant contrasts emerged.

In summary, these analyses provide additional evidence that the vmPFC encodes gaze-weighted sampled value signals and the pre-SMA encodes gaze-weighted accumulated value signals, though other correlations also emerged."

(2) The reporting of statistical results in the fMRI could be sharpened - e.g. in the figure legends, don't just say "Voxels thresholded at $p < .05$," but make clear whether you mean FWE whole-brain corrected (I think you do from the methods) or whether this is uncorrected for display; similarly, for the peak voxels, report the associated Z statistic at that voxel rather than just "negative beta".

We agree that it is important to include additional details regarding how we reported the statistical results. We now clarify our procedures in the main text:

"We report results using FWE-corrected statistical significance of $p < 0.05$ and a cluster significance threshold of $p < 0.005$."

We now also report the T statistics for peak voxels.

(3) A couple of the citations are slightly wrong - e.g., Kolling et al 2012 shouldn't be cited as arguing for decision conflict, as in fact it argues strongly against this account and in favour of a foraging account of ACC activity. Similarly, Hunt et al 2018 doesn't provide support for decision conflict; instead, it shows signals in ACC show evidence accumulation for left/right actions over time (although not whether these accumulator signals are gazewighted, in the same way as the present study).

We thank the reviewer for pointing out these mistakes in our citations. We have revised the references throughout.

Reviewer #2 (Recommendations for the authors):

(1) In some places, the introduction would benefit from fleshing out certain points. For example it is stated “For instance, decisions that are less predictable also tend to take more time (Konovalov & Krajbich, 2019) and can be influenced by attention manipulations (Parnamets et al., 2015; Tavares et al., 2017; Gwinn et al., 2019; Bhatnagar & Orquin, 2022). The quantitative relations between these measures argue for an evidenceaccumulation process.” It is not clear why the relations between them argue for an EA process, and the reader would benefit from some further explanation.

We thank the reviewer for this helpful suggestion. We agree that the original text did not sufficiently explain why these relationships support evidence-accumulation models. We have revised the introduction to better articulate the mechanistic basis for this claim.

This revision clarifies these points in the main text:

“Decisions like this are thought to rely on a bounded, evidence-accumulation process that depends on factors such as the value of the sampled information and shifts in attention. According to this framework, when two options are similar in value, evidence accumulates more slowly towards the decision threshold, resulting in longer response times (RT) and more opportunity for shifts in attention to influence the choice outcome. In contrast, when one option is clearly superior, evidence accumulates more rapidly and the decision is made quickly with less of a relation between gaze and choice. This choice process produces reliable, quantitative patterns in choice, RT, and eye-tracking data (Ashby et al., 2016; Callaway et al., 2021; Gluth et al., 2018; Krajbich et al., 2010; Smith & Krajbich, 2018). For instance, decisions with similar values are more random (i.e., less predictable), tend to take more time (Konovalov & Krajbich, 2019), and can be experimentally manipulated by diverting attention towards one option more than the other (Bhatnagar & Orquin, 2022; Gwinn et al., 2019; Parnamets et al., 2015; Pleskac et al., 2022; Tavares et al., 2017). Critically, these behavioral measures do not simply correlate; rather, they exhibit precise quantitative relationships consistent with evidence accumulation models (Konovalov & Krajbich, 2019).”

(2) Some of the study hypotheses also need to be clarified. What are the hypotheses regarding how SV and AV should translate to BOLD in an input vs integrator region? Larger SV/AV = larger BOLD? What predictions would be made for a time-on-task or conflict region? Are the predictions the same or different? Clarifying this will help the reader to understand to what extent the gaze manipulation is pivotal in identifying integrator regions.

We thank the reviewer for this excellent suggestion. We agree that it is useful to clearly articulate our hypotheses about BOLD signal predictions for different aspects of the model, and why gaze manipulation is critical for distinguishing between them. We have now expanded the introduction to clarify these predictions.

For input regions, we predicted a straightforward positive relationship: larger sampled value ($|\Delta SV|$) should produce larger BOLD activity. Input regions encode the momentary evidence being sampled (i.e., the relative value of currently presented stimuli). Consistent with prior work (Bartra et al., 2013), we expected such activity in the vmPFC and ventral striatum.

Critically, we also predicted that these sampled value signals should be modulated by gaze location. The attentional drift-diffusion model (aDDM; Krajbich et al., 2010) posits that attended items receive full value weight while unattended items are discounted. Consistent with prior work (Lim et al., 2011), we expected stronger vmPFC/striatum activity when the higher-value item is fixated compared to when the lower-value item is fixated

For integrator regions, we predicted an analogous positive relationship: larger accumulated value ($|\Delta AV|$) should produce more BOLD activity. Accumulator regions encode the summed evidence over the course of the decision. Consistent with prior work (Hare et al. 2011; Gluth et al. 2021; Pisauro et al. 2017) we expected such activity in the pre-SMA, dlPFC, and, IPS.

As with sampled value, we predicted that integrator activity should reflect gaze-weighted accumulated value. Just as inputs are modulated by current gaze, the accumulated evidence should be weighted by the history of gaze allocation over the entire trial.

Conflict-based models make qualitatively different predictions. Regions implementing conflict monitoring should show increased activity when options are similar in value, regardless of time.

The conflict account predicts that BOLD activity should scale with inverse value difference: smaller $|\Delta V| \rightarrow$ higher conflict $\rightarrow$ higher BOLD (Shenhav et al., 2014, 2016). In simple choice tasks, high conflict and high accumulated value are both associated with long RT (Pisauro et al. 2017), leading to ambiguity about how to interpret purported neural correlates of accumulated value. In our task we avoid this ambiguity – we analyze the effect of accumulated value at each point in time, not just at the time of decision. In this case, conflict should be inversely correlated with accumulated value. Moreover, the conflict account makes no predictions about how BOLD activity should be modulated by gaze allocation for a given set of values.

A more serious concern is the potential link to putative time-on-task BOLD activity. Accumulated value inevitably increases with time, leading to a correlation between the two variables (Grinband et al. 2011; Holroyd et al., 2018; Mumford et al. 2024). This is where the gaze data become particularly important. Time-on-task regions should show no relation with gaze allocation. After accounting for non-gaze-weighted accumulated value, only accumulator, and not time-on-task, regions should show a relation with gaze-weighted accumulated value. The results of the revised GLMs provide exactly such evidence.

We have edited the manuscript to make clear to readers why our gaze manipulation was not merely exploratory but rather a theoretically-motivated test to distinguish between competing models of decision-related neural activity.

We have clarified our study hypotheses in the Introduction as follows:

“We hypothesized that we would find (1) a positive correlation between gaze-weighted $|SV|$ and activity in the reward network (the ventromedial prefrontal cortex (vmPFC) and ventral striatum), and (2) a positive correlation between gaze-weighted $|AV|$ in the pre-supplementary motor area (pre-SMA) (Aquino et al., 2023), dorsolateral prefrontal cortex (dlPFC), and intraparietal sulcus (IPS).”

We have also added clarifying text about conflict and time-on-task to the Discussion as follows: “Conflict-based models make qualitatively different predictions. Regions implementing conflict monitoring should show increased activity when options are similar in value, regardless of time. The conflict account predicts that BOLD activity should scale with the inverse value difference: smaller $|\Delta V| \rightarrow$ higher conflict $\rightarrow$ higher BOLD (Shenhav et al., 2014, 2016). In simple choice tasks, high conflict and high accumulated value are both associated with long response times (Pisauro et al., 2017), leading to ambiguity about how to interpret purported neural correlates of accumulated value. In our task we avoided this ambiguity by analyzing the effect of accumulated value at each point in time, not just at the moment of decision. Under this approach, conflict should be inversely correlated with accumulated value (as higher accumulated evidence indicates less similarity between options). Moreover, the conflict account makes no predictions about how BOLD activity should be modulated by gaze allocation for a given set of option values.

A more serious concern is the potential confound with time-on-task BOLD activity. Accumulated value inevitably increases with time within a trial, leading to a correlation between the two variables (Grinband et al., 2011; Holroyd et al., 2018; Mumford et al., 2024). This is where the gaze data were particularly important. Time-on-task regions should show no relation with gaze allocation patterns. After accounting for non-gaze-weighted accumulated value, only accumulator regions, and not time-on-task regions, should show a relationship with gaze-weighted accumulated value. The results of our analyses provide exactly such evidence: preSMA activity was positively correlated with gaze-weighted accumulated value, even when accounting for previous gaze history and individual differences in attention discounting.”

(3) The authors allude to there being a correlation between SV and AV on this task, but the correlation is never reported. Please report the correlation with and without the removal of T-1.

We appreciate the reviewer pointing out this omission. We now report all correlations between SV and both the lagged and non-lagged versions of AV in the Methods section (Fig. 7). SV was significantly correlated with the full calculation of AV (Pearson’s $r = 0.27$). In contrast, this correlation, while still statistically significant, decreased when compared to lagged AV (Pearson’s $r = 0.06$).

(4) When examining relationships between SV, AV, and choice probability, the authors note that a larger coefficient for SV compared to AV is an inevitable consequence of an SSM choice process. Please explain why this is the case.

The reviewer is correct in observing that this point was not made sufficiently clear in the main text. We have now expanded the explanation in the behavioral results section.

The key insight is that in sequential sampling models, choices occur when accumulated evidence reaches a decision threshold. Importantly, the perceived value of each sample consists of the true underlying value plus random noise. The final sample (SV) is what pushes the accumulated evidence over the threshold, which creates a selection bias: decisions tend to occur when the noise component of SV happens to be positive and large. This means that the perceived final SV systematically overestimates the true SV, biasing upward the regression coefficient for the effect of SV on choice. In contrast, AV represents the sum of all previous sampled evidence, samples that we know did not lead to a choice. These samples are thus more likely to have had a negative or small noise component, meaning that the perceived AV systematically underestimates the true AV. This biases downwards the regression coefficient for the effect of AV on choice.

In the net, we expect that even when sample evidence is weighted equally over time in the true decision process, regression analyses will inevitably show larger coefficients for the effects of SV than for those of AV. This is a statistical artefact of the threshold-crossing mechanism, and not a reflection of differential weighting. We have incorporated this explanation into the revised manuscript to make clear why this pattern is an expected consequence of the SSM framework:

“The larger coefficient for ΔSV compared to ΔAV is an inevitable consequence of an SSM choice process. In SSMs, a choice occurs when accumulated evidence reaches a threshold. Critically, perceived value for any given sample consists of the true underlying value plus random noise. The final sample (ΔSV) is what pushes the accumulated evidence over the threshold, which creates a selection effect: decisions tend to be made when the noise component of ΔSV is relatively large and aligned with the ultimate choice, causing the perceived final ΔSV to systematically overestimate the true ΔSV . As a result, the regression coefficient for the effect of final ΔSV on choice is overestimated. In contrast, ΔAV represents

the sum of all previous evidence, which includes samples that were insufficient to trigger a choice and thus more likely to have noise components that favored the non-chosen option. This means that the perceived ΔAV systematically underestimates the true ΔAV . As a result, the regression coefficient for the effect of ΔAV on choice is underestimated. This creates an inherent asymmetry between ΔSV and ΔAV : even when the true decision process weights evidence equally over time, regression analyses will show larger coefficients for ΔSV than ΔAV . For any data generated by an SSM, regressing choice probability on final ΔSV and total ΔAV would produce a larger coefficient for ΔSV due to this threshold-crossing selection effect.”

(5) It is not clear to me why the authors single out the pre-SMA only in the abstract when IPS and dIPFC also show stronger correlations with AV and exhibit gaze modulation in the authors' final non-linear analysis. Further explanation is required in the Discussion and I would also suggest amending the Abstract because the 'Most importantly' claim will not be meaningful for the reader.

We appreciate the reviewer’s point. In the revised manuscript, we have included several new GLMs, including the new GLM1 that looks at gaze-weighted AV, above and beyond the effect of non-gaze-weighted AV. That analysis only supports pre-SMA. We have now clarified this in the Abstract as follows:

“Finally, we found gaze modulated accumulated-value signals, above and beyond the non-gazemodulated signals, in the pre-supplementary motor area (pre-SMA), providing novel evidence that visual attention has lasting effects on decision variables and suggesting that activity in the pre-SMA reflects accumulated evidence.”

(6) Some discussion of statistical power would be warranted given that a sample of 23 is now considered small by current fMRI standards.

We appreciate the reviewer raising this important issue. We acknowledge that our sample size of 23 subjects (with only 20 having useable eye-tracking data) is on the small side by current fMRI standards. However, we believe several features of our study design and analytic approach mitigate concerns regarding statistical power.

First, our paradigm leveraged a within-subjects design with high total sample counts. Each participant completed approximately 60 choice trials across three 15-minute runs, with an average of 6.37 samples per trial. This yielded roughly 380 observations per participant, providing substantial statistical power at the individual level before aggregating across subjects. This within-subject power is particularly important for detecting parametric effects, as our regressors of interest ($|\Delta SV|$ and $|\Delta AV|$) varied continuously across and within trials.

Second, rather than conducting an exploratory whole-brain analysis that would require larger sample sizes to correct for multiple comparisons, we employed a targeted ROI approach based on well-established regions from prior literature (e.g., Bartra et al., 2013; Hare et al., 2011). This ROI-driven approach substantially increases statistical power by reducing the search space and leverages theoretical predictions about where effects should occur. Our novel contribution that gaze modulation of accumulated evidence signals was reflected in pre-SMA activity builds naturally on established findings.

However, we acknowledge that a larger sample size would provide greater confidence in the null effects and would enable more detailed individual differences analyses.

We have added a brief acknowledgement of the sample size limitation to the Discussion section of the main text:

“While our sample size of 20 subjects is modest by current neuroimaging standards, the withinsubject statistical power from our extended decision paradigm (~380 observations per

subject), combined with hypothesis-driven ROI analyses and multiple comparisons correction, provides confidence in our core findings. Nevertheless, replication with larger samples would be valuable, particularly for more fully characterizing null effects and marginal findings.”

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